

Experience Effects in the Reception of Political News: Evidence from Trump's Miss USA Pageant*

HENG CHEN

University of Hong Kong

YANG XU

University of Hong Kong

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Abstract. Does a voter's past determine which news she takes in? We study the 2016 *Access Hollywood* tape, which tied Donald Trump to the Miss USA pageant he owned for two decades. Exposure to the pageant is assigned by county and birth year: whether a resident was old enough to follow a local titleholder crowned under his ownership. A difference-in-differences analysis shows that where exposure was higher, campaign contributions to Trump fell by more from the day of the release, intentions to vote for him fell by more in a daily survey, and his vote share was lower in November and in 2020. Further evidence and a simple model point to the mechanism: memory of the title raised attention to the news, and voters revised their assessment of Trump. The findings speak to negative campaigning and to the demand for news.

Keywords: political accountability; information; attention; memory; experience effects; campaign contributions; scandal.

JEL Classification. D72, D83, D91, L82.

*Heng CHEN: The Faculty of Business and Economics, The University of Hong Kong; Email: hengchen@hku.hk. Yang XU: The Faculty of Business and Economics, The University of Hong Kong; Email: xyethan@connect.hku.hk. This paper is a substantially revised version of, and supersedes, an earlier draft circulated under the title "Pageants and the President: The Liability of Celebrity Politicians".

1. Introduction

Democratic accountability rests on voters learning about their politicians. A large literature has established that when voters are given information about misconduct—audit reports, disclosures, media coverage—they punish it at the polls, and that where such information is scarce or its supply is obstructed, accountability weakens.¹ This work treats reception as following from supply: information that reaches a voter is information she has. Yet voters in modern democracies live in an information-rich environment. They face more political news than they can attend to; some of it registers and shapes their political behavior while the rest passes them by. Which news voters attend to, and why, therefore has political consequences of its own.

Existing work on the demand for news points to three motives: a concern for accuracy, a desire to see one’s own views confirmed, and the value of the news for the decisions the reader faces.² The first two are properties of the content, how reliable it is and which side it favors; the third is a property of the reader’s situation, what the news is worth to her. This paper studies a determinant of a different kind, a property of the reader herself: her own past. Studies of experience effects show that lived history shapes what people believe and what they want.³ If that is the case, it plausibly also shapes which news a voter attends to, and how closely. The experience we study is a personal or local connection to the subject of the news, the kind that makes a national story about a politician also a story about one’s own town, school, or workplace.⁴ We ask whether voters respond more strongly to news about a politician when they already have such a connection to its subject. If they do, the same disclosure can move some voters and leave others untouched, and the distribution of such connections across the electorate becomes a determinant of accountability in its own right.

Testing this conjecture faces two obstacles. The first is identification. A connection to the subject of the news is rarely separable from political preferences, so that any differential response could reflect partisanship or confirmation rather than the connection itself. The second is measurement. Even where such a connection exists, it is seldom

¹Ferraz and Finan (2008, 2011) show that the public release of municipal audits in Brazil reduced the re-election of mayors found to be corrupt; Kendall et al. (2015) that an information campaign in an Italian mayoral election moved voters’ beliefs and votes; and Arias et al. (2022) that the electoral effect of a malfeasance revelation depends on what voters expected of the incumbent.

²See Gentzkow and Shapiro (2006, 2010) on accuracy and slant, Chopra et al. (2024) for a direct test of the first two motives against each other, and Prat and Strömberg (2013) for a survey of the field and Roth et al. (2022) on the demand for news that bears on one’s own exposures.

³Those who grew up in a recession take less risk and those who lived through high inflation expect more of it decades later, while those who lived under communism hold different views of redistribution (Malmendier and Nagel, 2011, 2016; Alesina and Fuchs-Schündeln, 2007).

⁴Recent evidence from macroeconomics points the same way: the response to a macroeconomic shock is concentrated among people whose past contains a similar episode (Link et al., 2026; Gennaioli et al., 2024).

observed for individual voters.

The setting we study overcomes both obstacles. On October 7, 2016, one month before the presidential election, the *Washington Post* released a 2005 recording in which Donald Trump described, in lewd terms, his conduct toward women. The story was available to every voter on the same day and through outlets of every political leaning.⁵

The content of the tape reached deep into Trump's pageant past. The woman whose treatment Trump described, Nancy O'Dell, was a Miss USA alumna who had worked with Trump during his ownership, and the coverage of the tape revived his separate remarks, from a 2005 radio interview, about entering contestants' dressing rooms as the pageant's owner. In the coverage that reached voters, the scandal was thus tied to the nationally televised Miss USA pageant, which Trump personally owned for nearly two decades (1996–2015). State titleholders, crowned in independently licensed state-level contests, all advanced to Trump's national final: when a woman from a given county appeared on that stage, residents of her home county were more likely to follow the pageant and to encounter Trump as its owner.

This tie is the source of our variation. Whether a voter was likely to have formed a prior connection to the institution at the center of the scandal depends on whether her county produced a titleholder during Trump's ownership and on whether she was old enough at the time to follow it—on where she lived and when she was born, not on her politics. The connection was formed years before the campaign, through an entertainment institution rather than a political one, which separates it from partisanship and confirmation; and because its likelihood is determined by county and birth year, it can be measured for every voter.

We measure it from the hometowns of every Miss USA state titleholder crowned between 1952 and 2015, which we map to 713 counties. For each county we construct an exposure share: the fraction of its voting-eligible population that was at least ten years old when a titleholder from the county was crowned during Trump's ownership. The share assigns, from county and birth year, the opportunity to have followed a local title. To mitigate potential selection issues, our baseline sample is the set of counties that have ever produced a titleholder, so that comparisons are made among pageant-producing communities rather than between them and communities that never produced one. Within this sample the exposure share varies with the timing of titles and the age structure of the population, and balance tests show that none of a rich set of county characteristics predicts it.

⁵The release set off one of the largest media firestorms of the campaign. Within days Trump issued a rare recorded apology; by October 11, roughly a third of sitting Senate Republicans had said they would not vote for him, and House Speaker Paul Ryan announced he would no longer publicly defend his candidacy; national polls placed him as much as fourteen points behind Clinton by mid-October (Dann, 2016).

Our principal outcome is campaign contributions. Contributions are a costly expression of support: a person who gives to a candidate reveals a preference in a way that a survey answer does not, and individual giving behaves like participation rather than investment, tracking the motives that drive turnout and vote choice (Ansolabehere et al., 2003; Bonica, 2014; Barber et al., 2017). They are also recorded to the day: from the Database on Ideology, Money in Politics, and Elections (DIME) we build a daily panel of contributions for each of the 713 sample counties from September 1, 2016 to election day.

We estimate a difference-in-differences specification that compares the change in contributions before and after October 7 across counties with different exposure shares, with county fixed effects absorbing all time-invariant differences among counties and date fixed effects absorbing the national course of the campaign. The comparison holds supply fixed and isolates the role of the connection in what voters took in. We ask whether the same news met a different response where the connection was stronger; the five weeks of daily data before the release allow the parallel-trends assumption on which it rests to be examined directly.

After the release, contributions to Trump fell by more in counties with a higher exposure share. A unit difference in exposure share—about two standard deviations—lowers the inverse hyperbolic sine of contributions by 0.54 units, or about 20 percent of the sample mean. The decline appears on both margins. The probability that a county records any contribution to Trump on a given day falls by 3.8 percentage points from a base of 36 percent, and a unit difference in exposure share corresponds to about two fewer contributions and roughly one fewer distinct contributor per day, declines of about three-quarters of the respective means, whereas the amount per contribution falls by only about one-sixth. The decline is thus concentrated on the extensive margin of giving, in the number of contributions and of contributors rather than in the amount per contribution. An event-study specification shows no differential movement over 36 days before the release, and a decline that begins immediately after it and persists through election day.

The response is specific to Trump and to the connection. Contributions to other Republican candidates and to Clinton show no differential change with the exposure share: previously exposed communities neither turned against the party nor redirected their giving to the opponent. An exposure share built from Miss America titleholders—a comparable national pageant that Trump never owned—is unrelated to contributions to Trump after the tape, so the response does not reflect some unobservable feature of communities with a pageant tradition.

To complement the evidence across counties, we provide evidence within counties

at the individual level, using the Cooperative Congressional Election Study (CCES), a survey fielded daily through the campaign. The comparison is between residents of the same county, interviewed on the same day, who differ in whether they were old enough to have followed a local title; county-by-survey-date fixed effects absorb every local shock, including local coverage of the tape. A unit of exposure lowers the probability of intending to vote for Trump after the release by about three percentage points, against a mean of one-third. There is no differential movement before the release, and exposure bears no relationship to the primary vote cast months earlier.

We then ask whether the response reached the ballot box. The vote is the most consequential downstream outcome, and the contribution evidence alone cannot tell whether the withdrawal of support was confined to donors, or to the weeks after the release. In the 2016 election, a unit difference in exposure share is associated with a Trump vote share 0.8 percentage points lower, conditional on state fixed effects, the county's previous Republican vote share, and demographic and economic controls. Conditional on the previous election's Republican vote share, the negative association is 0.5 percentage points in 2020 and is small and statistically insignificant in 2024. We find no statistically significant association with the 2012 Romney vote or with aggregate turnout in 2016; the latter result does not rule out offsetting changes in participation across voter groups. Election returns offer no timing variation within the campaign, so beyond the 2012 placebo we guard against pre-existing differences across counties in a number of ways.

We then look inside the response, following the three links through which a connection formed years earlier could shape the response to news: whether previously exposed voters attended to the tape, whether the response depends on how recently the connection was formed, and what voters revised. On attention, the evidence suggests that after the release, search interest in the tape and in Trump rose by more in markets containing counties with higher exposure shares, while search for the other Trump story of the campaign did not, and the Miss America share predicted none of these. In the survey, the decline in support is concentrated among respondents who follow political news, although the connection did not change whether they did. On memory, the point estimates grow with the recency of the connection at both levels: the contribution response and individual vote intentions. On updating, previously exposed voters became less likely to support Trump without changing where they placed him ideologically, a pattern consistent with reassessing the candidate rather than his positions; the response did not extend to Republican candidates down the ballot, and it was not concentrated in any demographic group.

Taken together, the three sets of results support a single inference: a connection formed outside politics, years before the campaign, changed how voters responded to

the same news, and it did so in behavior that was costly and reached the ballot box.

Two takeaways follow. The first bears on a puzzle in the study of campaigns. Negative campaigning is close to universal, yet the research record finds that its average effects are small and inconsistent (Lau et al., 2007; Ansolabehere and Iyengar, 1995; Geer, 2006). Our findings suggest one reason. Damaging information about a candidate's past does not land uniformly; it lands where voters carry a connection to what it is about, and are more likely to pay attention to it. An effect that is localized in this way is easy to miss in an average and hard to miss for a campaign that targets. The point matters more now than it once did: candidates increasingly enter politics from outside it, carrying long non-political histories with wide but uneven reach.

The second concerns how political news takes effect. The prior connection to the scandal changed neither the reliability of the news nor its slant nor its value for the vote; what it plausibly changed is the cost of taking the news in, through the reader's own past, and the attention evidence, though indirect, points that way. Reception is thus a margin of its own, and it can vary across places and generations for reasons that have nothing to do with ideology, so that two voters with the same politics and the same media can take in the same story differently.

We propose a simple model of news filtering to organize these facts and to characterize the mechanism. A voter wants an accurate belief about a candidate's fitness for office. An event occurs, and its cue—the people and the institution it involves—acts on her memory, bringing past experience to mind in proportion to its similarity to the cue and its recency. The share of recall the cue captures is her connection to the event, and it lowers the cost of attending to the news; she then chooses how much attention to pay, as in models of endogenous information acquisition, and her belief is updated. The model yields four predictions as comparative statics. A stronger connection moves a voter from ignoring the news to attending to it, raises attention conditional on attending, and produces a larger revision of her assessment; and the response is weaker the older the experience behind the connection. Each has an empirical counterpart: the extensive margin of the contribution response, the rise in search for the tape but not for other Trump news, with no rise in news consumption at large, the contrast between support and ideological placement, and their specificity to the person the cue named, and the recency gradient.

This paper contributes to several literatures. First, it extends the literature on experience effects to the acquisition of news and attention to it, and traces the political consequences in campaign contributions and votes. That literature shows that lived history shapes what people believe and what they want, directly and for decades (Malmendier and Nagel, 2011, 2016; Alesina and Fuchs-Schündeln, 2007; Malmendier,

2021). Our findings suggest that it also shapes what people take in: a connection formed years earlier, outside politics, determined which voters attended to a national story and how much they revised in response.

Second, our paper contributes to the literature on the demand for news. This literature emphasizes readers' concerns for accuracy and confirmation and their implications for media slant (Mullainathan and Shleifer, 2005; Gentzkow and Shapiro, 2006, 2010; Chopra et al., 2024); Strömberg (2015) surveys the field. It also documents substantial differences in news consumption and knowledge across readers. Prior (2007) shows that greater media choice allows preferences for news versus entertainment to translate into wider gaps in political knowledge and participation, while Angelucci and Prat (2024) document substantial socioeconomic differences in voters' ability to distinguish important political news from false stories. We identify a further source of variation in news reception: the connection between a story and the reader's prior experience. Such a connection can shape whether a voter attends to a particular story and how closely, even among those who already follow political news.

Third, our evidence on the mechanism is consistent with the view that people are more likely to attend to news whose cues bring similar and recent experiences to mind. This interpretation builds on work on associative memory (Bordalo et al., 2020, 2023; Enke et al., 2024; Wachter and Kahana, 2024; Graeber et al., 2024) and on evidence that responses to macroeconomic shocks are concentrated among those who have experienced similar episodes (Link et al., 2026; Gennaioli et al., 2024). In the latter studies, past experience and current news concern the same subject, so recall and a personal stake in the news go together. In our setting, the experience and the news share a feature, but the experience gives the voter no stake in the outcome of the news. This distinction helps separate the memory channel from the value of the information. Section 7 formalizes how an event's cue activates memory and how memory, in turn, affects the cost of attention, drawing on the tradition of rational inattention.

Finally, our setting connects to work on celebrity politics and the electoral value of fame acquired outside politics (West and Orman, 2003; Wheeler, 2013), as well as evidence that celebrity endorsements and entertainment exposure can influence voting (Garthwaite and Moore, 2013; Durante et al., 2019; Kim and Patterson Jr, 2025). The closest study is Xiong (2021), who finds that exposure to Ronald Reagan as host of *General Electric Theater* increased subsequent support for his candidacy—a celebrity premium. His focus is on the electoral advantage conferred by prior influence; ours is on how that exposure shapes voters' attention to, and response to, subsequent news about the candidate.

2. Background

This section establishes three premises on which the design rests. The Miss USA pageant is where a connection to Trump was formed, and that connection was local, dated, and formed outside politics (Section 2.1). The *Access Hollywood* tape was national news whose content pointed at that connection (Section 2.2). And the tape reached voters with such a connection as unfavorable news about Trump, not as an attack on an institution they were attached to (Section 2.3).

2.1. Miss USA and the Local Connection to Trump

Miss USA is the institution through which the connection we study was formed. Three properties of that connection matter for the design. It was local: titles were won by women from particular communities. It was dated: each title was an event of a particular year. And it was formed outside politics: through an entertainment institution, years before Trump's candidacy. This subsection documents the three, and presents evidence that a local title drew a community's attention to the pageant and, through it, to Trump as its owner.

The pageant was established in 1952 and operates in two stages. First, each of the fifty states and the District of Columbia holds a preliminary competition, licensed to local directors, to crown a "Miss [State] USA"—the state titleholder. Second, every state titleholder advances to the nationally televised final. A title is thus a hometown event with a national stage: the contest that produces it is local, and the stage it leads to is the same for every state.

Trump acquired the Miss Universe Organization, which includes Miss USA, in 1996 and owned it until September 2015, three months into his presidential campaign. Over those two decades the pageant was among the most visible parts of his public life before politics, and it reached communities that had no political contact with him: the finals aired on national television, in primetime from 2002, and each year's edition carried a titleholder from every state.

Following the pageant meant encountering Trump as its owner. He appeared on stage at the final ceremony, and winners' acceptance remarks frequently credited him by name; Alyssa Campanella (Miss USA 2011), for example, said in hers that the opportunity "could not have been possible if it weren't for Mr. Donald Trump." Broadcast nationally and replayed in the winner's home community, such moments made Trump part of what a local title brought to a community's attention.

A connection to the pageant was therefore local, dated, and non-political. Each time a woman from a county won the Miss [State] USA title and advanced to the national

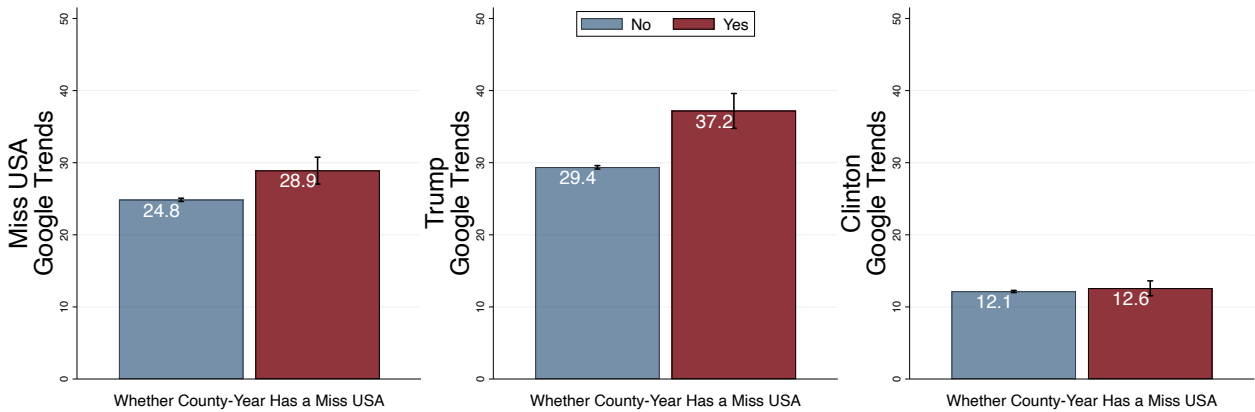


Figure 1. Google Trends for “Miss USA,” “Trump,” and “Clinton” in Miss USA State Titleholder’s Hometown

Notes: This figure plots the Google Trends search interest for the terms “Miss USA,” “Trump,” and “Clinton” in a county-year panel from 2004 to 2015, with each county assigned the search interest of its DMA. Whether County-Year Has a Miss USA indicates whether the county produced a Miss USA state titleholder in that year.

stage, residents of her community were more likely to follow the pageant that year and, in doing so, to encounter Trump as its owner. The connection was formed where and when a title was won, and renewed when another was; and it was formed through an entertainment institution, years before Trump entered politics, so that whether a voter was likely to carry it depended on where she lived and when she was born rather than on her political preferences; it is these properties that the exposure measures of Section 4 exploit.

We illustrate this local connection using Google Trends data. Because Google Trends data are available at the designated market area (DMA) level—where each DMA aggregates the counties sharing a common primary television market—we match each state titleholder’s home county to its corresponding DMA and measure search volume in that area during her crowning year. Figure 1 reports the results for three search terms. A Miss USA victory is associated with elevated “Miss USA” search interest in the winner’s DMA, confirming that the search data capture local attention to the pageant. The same victory is associated with a rise in “Trump”-related search activity, linking the pageant to Trump-specific local visibility. By contrast, we find no comparable pattern for “Clinton,” ruling out a general rise in political attention. A local title thus brought Trump, and not politics in general, to a community’s attention; the sections that follow ask what this connection did when the tape arrived.

2.2. The Access Hollywood Tape

On October 7, 2016, approximately one month before the presidential election, The Washington Post published a leaked 2005 video recording (the “Access Hollywood” tape) in which Donald Trump described, in lewd terms, his conduct toward women. The story led every national broadcast and every major newspaper for days, was replayed at the presidential debate two days later, and was carried by outlets of every political leaning at once.

The discourse surrounding the tape directly linked Trump’s personal conduct to his professional history as the pageant’s proprietor. The woman discussed in the recording, Nancy O’Dell, was herself a Miss USA pageant alumna—Miss South Carolina USA 1987 and a co-host of both Miss USA and Miss Universe from 2004 to 2007 who had worked directly with Trump during his ownership. The coverage also revived Trump’s separate remarks, from a 2005 interview with Howard Stern, in which he had tied his ownership to his access to contestants: “I’m allowed to go in [the dressing rooms] because I’m the owner of the pageant.” The pageant was thus part of what the tape was about: the story as voters encountered it pointed at the institution, and a voter who heard of the tape heard of the pageant with it.

The public read the tape this way. Figure 2 shows that search interest for “Miss USA” rose in step with searches for “Access Hollywood” immediately after the release, so the pageant was drawn into the coverage of the tape rather than remaining in the background. For a voter with a connection to the pageant, the story therefore carried a feature that was hers in particular. Whether that feature changed how she took in the news is the question of the rest of the paper.

2.3. The Tape as Unfavorable News

Did the tape reach previously exposed voters as unfavorable news about Trump? Nationally there is little doubt: Trump’s polling fell behind Clinton in the days after the release (Sides et al., 2018; Rhodes et al., 2020), as one expects for a story that dominated the news for days. The question is whether the same was true of voters with a prior connection to the pageant. The design treats the connection as separate from political preferences because it was formed outside politics. A skeptic may reply that following a pageant Trump owned could have produced affinity for him, so that among exactly these voters the tape could have been discounted or met a defensive response; if so, a differential response among them could not be read as a response to unfavorable news. The evidence in this subsection speaks to that alternative.

We use YouTube comment data to provide some suggestive evidence. We collect

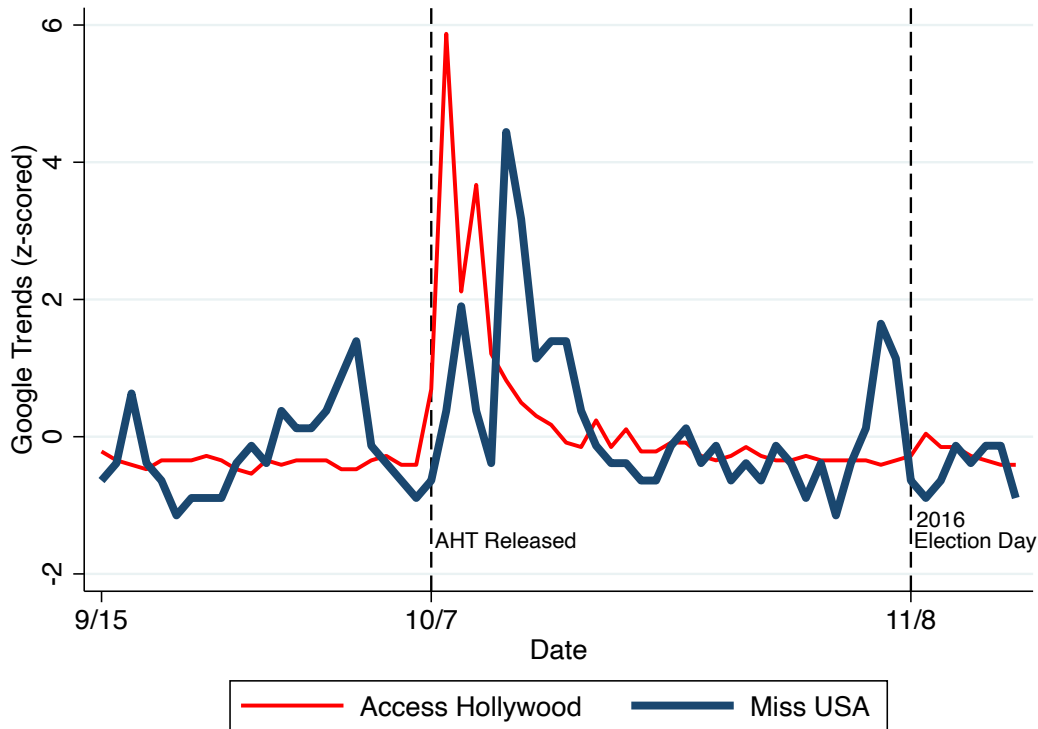


Figure 2. Google Trends for “Access Hollywood” and “Miss USA”

Notes: This figure plots the Google Trends search interest for the terms “Access Hollywood” and “Miss USA” from Sep 15 to Nov 15, 2016. The search interest is z-score normalized.

user comments from the livestreams of the 2016 presidential debates on the NBC News YouTube channel using the YouTube API, and match these commenters to Miss USA viewers through user identifiers obtained from the official Miss USA YouTube channel. As a comparison group, we collect comments from the American Idol and The Voice official YouTube channels and link debate commenters to these programs through the same user identifiers. This exercise identifies voters with a prior connection by their following of the pageant online rather than by a hometown title; it is a different and coarser proxy for the same connection, which we use because commenters’ hometowns cannot be observed.

Before the tape’s release, Miss USA viewers’ debate comments exhibited no discernible difference in sentiment toward Trump relative to other commenters, and the same was true for the comparison programs. After the leak, the pattern diverged: Miss USA viewers expressed more negative sentiment toward Trump, while viewers of American Idol and The Voice did not show a comparable shift. Appendix Figure A1 reports the underlying coefficients. We do not interpret this exercise as a rigorous test; we read it as suggesting that the tape reached those with a prior connection to the pageant as unfavorable news about Trump, and more so than it reached others. In

other words, the connection did not shield Trump; it did the opposite.

3. Data

This section describes the data by their role in the analysis. The titleholder data measure the connection: where and when a local title was won (Section 3.1). Campaign contributions, election returns, and the 2016 CCES measure the response, at the county-day, county, and individual level (Sections 3.2–3.4). Search data and one CCES item measure what voters took in (Section 3.5). Section 3.6 defines the baseline sample and the comparison samples.

3.1. Miss USA and Miss America State Titleholders

We assemble a comprehensive dataset of Miss USA state titleholders from 1952 to 2015, drawing primarily on publicly available records compiled in Wikipedia entries, which themselves draw on *Newspapers.com* and other archival sources. For each titleholder, we collect city-level hometown information—the community she represented at the time of the title—and match her to a county of origin, yielding a dataset of 2,594 state titleholders from 713 unique hometown counties.⁶ Figure B1 displays the geographic distribution of Miss USA state titleholders across U.S. counties. Each title is thus tied to a county and a year. The titles crowned during Trump’s ownership, 1996–2015, are the ones that enter the exposure measures of Section 4.

We construct a parallel dataset for Miss America, a national pageant of the same two-stage structure that Trump never owned, by the identical procedure, yielding 2,771 state titleholders from 887 counties, of which 430 coincide with a Miss USA titleholder from the same county in the same year. Section 3.6 describes how Miss America counties—communities with a pageant tradition but no connection to Trump—serve as a placebo and as a comparison group.

3.2. Campaign Contributions

Our second source, and the source of our principal outcome, is the Database on Ideology, Money in Politics, and Elections (DIME), which records the amount, date, location, and recipient of individual campaign contributions reported to the Federal Election Commission (Bonica, 2014). We aggregate contributions to a county-by-day panel for the 713 sample counties over the period from September 1, 2016 to election day, November 8, 2016, which spans five weeks before and four and a half weeks after the release of the tape on October 7. For each county-day we observe total contributions

⁶The shortfall relative to the upper bound ($51 \text{ jurisdictions} \times 64 \text{ years} \approx 3,260 \text{ entries}$) reflects incomplete early rosters: in the inaugural 1952 competition only 30 delegates competed, and the pageant grew organically through the 1950s and 1960s.

to Trump, to other Republican candidates, and to Hillary Clinton, together with the number of contributions and of distinct contributors.

Two features of these data serve the design. The daily resolution lets the response be dated to the release itself, so that the timing of the response can be compared with the timing of the news. And the counts of contributions and of contributors separate the extensive margin of giving—whether anyone in a county gives on a given day—from the amount given, which is the margin on which Section 5.1 finds the response. Contributions in a county-day cell are highly skewed and nearly two-thirds of cells are zero: the mean contribution to Trump is \$842 per county-day, from 2.6 contributions and 1.3 distinct contributors on average, and 36 percent of county-days record at least one contribution to him.

3.3. Election Returns and County Characteristics

County-level presidential election returns for 2012–2024 are obtained from the MIT Election Data and Science Lab’s *County Presidential Election Returns*. We augment these data with a rich set of socioeconomic and demographic controls, including population size, racial composition, gender and age distributions from the 2016 American Community Survey (ACS); payroll and employment statistics from the County Business Patterns; and income and poverty measures from the Small Area Income and Poverty Estimates (SAIPE). The county exposure share of Section 4.1 requires the age distribution of each county’s voting-eligible population, which we take from the same source; it is the age distribution that converts a title won in a given year into the share of a county’s residents who were old enough to follow it.

Appendix Table B1 reports county-level summary statistics for the main analysis sample of 708 counties.⁷ The average Republican vote share is 55.5% in 2016, with comparable levels in prior presidential elections. Counties in the sample have an average poverty rate of 14.4% and a median household income of approximately \$55,000. On average, counties have a population of 0.30 million, with Black, Hispanic, and Asian population shares of 10.3%, 9.3%, and 2.7%, respectively.

3.4. The CCES Survey

To look inside the response—at what individual voters revised, and who among them did so—we use data from the 2016 Cooperative Congressional Election Study (CCES). The CCES follows a two-wave design: a pre-election wave fielded *daily* between

⁷Alaska’s returns are reported by state house district rather than by borough or census area, so the five Alaska counties that produced a titleholder—Anchorage, Fairbanks North Star, Kenai Peninsula, Southeast Fairbanks, and Aleutians West—have no county-level vote share and are excluded from the electoral specifications, leaving 708 of the 713 titleholder counties. The District of Columbia is absorbed by the state fixed effects, so the electoral regressions are estimated on 707 counties.

September 28 and November 7, 2016, and a post-election wave completed by 52,899 of the 64,600 initial respondents after Election Day. The pre-election wave covers 64,600 respondents from 2,668 counties, of whom 16,990 were interviewed prior to the release of the Access Hollywood tape on October 7 and 47,610 thereafter. The CCES provides detailed information on voting behavior, political attitudes, and demographic characteristics, including respondents' county of residence, birth year, interview date, gender, education, race, employment status, party registration, social media usage, and ideological self-placement.

The pre-election wave elicits respondents' immediate presidential vote preferences. Among those who report an intention to vote and choose between Donald Trump and Hillary Clinton, 16,741 out of 40,573 express support for Trump. It also asks respondents to place Trump on a seven-point ideological scale; Section 6.3 uses support and placement together, since a change in the first without a change in the second distinguishes a reassessment of the candidate from a reassessment of his positions. The survey also asks whether respondents voted for Trump in the primary elections, which we use as a placebo test. Finally, it asks whether the respondent had recently read a story or watched a video about politics, which we use in Section 6.1 as an indicator of whether she was following political news—the individual-level measure of intake.

The post-election wave, completed after Election Day, records respondents' reported general-election votes. We use it only in Appendix F, which shows that respondents with greater exposure were less likely to report having voted for Trump in 2016, with no corresponding relationship for the 2012 vote.

Appendix Table B2 reports summary statistics for the CCES individual-level sample. The sample includes 40,573 respondents, of whom 41.3% report a preference for voting for Donald Trump. The average birth year is 1966, and 48% of respondents are male. Roughly half hold a four-year college degree, and 72.9% are White. The remaining characteristics—employment status, party registration, social media use, and ideological self-placement—enter the individual specifications as controls, and their distributions are reported in the appendix table. As at the county level, the baseline individual sample is respondents in the counties of the baseline county sample; the full sample is reported alongside it.

3.5. Search Data

Our fifth source is Google Trends, which reports search interest by designated market area (DMA), the television market that groups neighboring counties; we match each county to its DMA. We use the data in two forms. In Section 2, search interest for “Miss USA,” “Trump,” and “Clinton” in each titleholder's DMA in her crowning year, over 2004–2015, shows what a local title brought to a community's attention. In Section 6.1,

a county-week panel measures attention to the tape. Because Google Trends returns one value per market and query, we query each DMA separately for each of the ten seven-day windows from September 1 to November 8, 2016, and assign each county the search interest of its DMA in that week, weighted by the county's share of the DMA population. The panel has one observation per county and week, for the terms "Trump," "Access Hollywood," and "Russian interference." Search interest in the tape is the paper's direct measure of attention; because it varies across markets rather than counties, the attention results are read as comparisons across DMAs.

3.6. Miss USA and Miss America Counties

A natural question is whether counties that produced Miss USA state titleholders differ systematically from the rest of the country. Appendix Table B3 reports linear probability regressions in which the dependent variable is an indicator for whether a county produced a Miss USA state titleholder. Column (2) compares Miss USA counties to all other U.S. counties, including state fixed effects. Several county characteristics are significant predictors: Miss USA counties tend to have higher median household incomes, greater employment levels, a younger age profile, and larger Asian, Black, and Hispanic population shares. These patterns indicate that Miss USA counties are not a random draw from the national distribution; they differ from the broader population of U.S. counties in ways that could correlate with political outcomes.

This selection concern motivates our sample restriction. Comparing Miss USA counties to the full national sample would conflate the effect of the connection with pre-existing differences between pageant and non-pageant counties. We therefore restrict the baseline sample to the counties that have ever produced a Miss USA state titleholder, which removes the comparison between communities with and without a pageant tradition; what remains is variation in when titles were won and in how old residents were at the time. Within this set, identification comes from the two sources of variation described in Section 4: across counties, in the share of the voting-eligible population that was old enough to follow a title won during Trump's ownership; and within counties, across birth cohorts, in whether a respondent was old enough to follow such a title when it was won. We report results using the full national sample only as a robustness check.

Miss America counties provide a second comparison. Conditional on being a pageant-producing community, Miss USA and Miss America counties are observationally equivalent: none of Republican vote share, poverty rate, income, employment, population, or racial composition distinguishes them (Appendix Table B3, column 1). We use them in two ways. We construct a Miss America exposure share by the recipe of Equation (1), from Miss America state titleholders crowned during 1996–2015, for use

as a placebo treatment. And we form a pooled sample of Miss USA and Miss America counties, together with a matched sample in which Miss America counties are matched to Miss USA counties on observable characteristics by propensity score, for use as a comparison group.⁸

4. Empirical Design

The prior exposure to the event subject is at the center of the paper, a voter’s prior connection to the Miss USA pageant, and through it to Donald Trump as its owner. Within a county, residents of different birth cohorts were at different ages when a local titleholder appeared, and only those old enough to follow the pageant were in a position to form a connection to it. The release of the tape on October 7, 2016 supplies the before-and-after comparison, which allows us to compare the change in an outcome around the release across voters or counties with different prior exposure—a difference-in-differences design.

We implement it at two levels, matching the two sets of outcomes: a county level exposure share for campaign contributions, election returns, and an individual exposure measure for survey respondents, built in parallel and permitting comparisons of cohorts within the same county.

4.1. Campaign Contributions: County Exposure Share and Difference-in-Differences

Sample and exposure. The baseline sample is the set of counties that have ever produced a Miss USA state titleholder; we report other samples where stated. All comparisons are therefore among pageant-producing communities, and the variation we use comes from the timing of titles relative to Trump’s ownership (1996–2015) and relative to residents’ ages, not from whether a community produced a titleholder at all. We treat a resident as old enough to have followed the pageant if she was aged 10 or older at the time of the title, following the World Health Organization’s definition of the onset of adolescence; young children are unlikely to follow or be aware of a beauty pageant, and Section 5.2 varies this threshold. Both exposure measures assign, on the basis of county and birth year, the opportunity to have followed a local title.

For each county c and birth cohort b , let $\mathbb{1}(Exposed_{cb})$ equal one if cohort b was aged 10 or older when a Trump-era Miss USA state titleholder from county c was crowned,

⁸We employ 1:1 nearest-neighbor matching with a caliper of 0.01 on the previous Republican vote share, poverty rate, median household income, employment, annual payroll, number of firms, population, youth ratio, sex ratio, and the shares of Black, Asian, and Hispanic residents. The matched sample contains 887 counties (625 Miss USA and 262 Miss America); the mean standardized bias falls from 32.1% to 2.6%, the pseudo- R^2 from 0.099 to 0.002, and a joint balance test does not reject ($p = 0.986$). Appendix Figure C3 plots the propensity-score distributions.

and let $\mathbb{1}(Eligible_b)$ equal one if cohort b had reached voting age by November 2016. The county exposure share is the fraction of the county's voting-eligible population that was in a position to be exposed:

$$\text{Exposure Share}_c = \sum_b \mathbb{1}(Exposed_{cb}) \cdot \mathbb{1}(Eligible_b) \times \frac{Pop_{cb}}{Pop_c}, \quad (1)$$

where Pop_{cb} and Pop_c are the cohort and total county voting-age populations, so that cohorts are weighted by their share of the electorate. The eligibility condition matters because the county measure is built from the full population across all birth cohorts, and it aligns the share with the electorate whose response the outcomes measure.

Balance tests. Since every county in the sample has produced a titleholder, counties do not differ in pageant participation; the exposure share varies with whether a county's titleholder was crowned early or late in Trump's ownership and with how many of its cohorts were old enough at the time. Such timing is plausibly unrelated to underlying county characteristics. Regressing a rich set of county political, socioeconomic, and demographic characteristics, including the previous election's Republican vote share, on the exposure share shows no systematic relationship (Appendix Figure C1), and an indicator for producing a titleholder in the early Trump era (1996–2005), conditional on state fixed effects and the total number of titleholders, is likewise unrelated to county characteristics (Appendix Figure C2).

Specifications. The county-level analysis is a difference-in-differences design. For daily campaign contributions, we compare the change in an outcome before and after the release of the tape across counties with different exposure shares:

$$y_{ct} = \beta \text{Exposure Share}_c \times \text{Post}_t + \theta_c + \delta_t + \epsilon_{ct}, \quad (2)$$

where y_{ct} is the outcome in county c on date t , Post_t indicates dates after the release, and θ_c and δ_t are county and date fixed effects, so that β is identified from the differential change around October 7 across counties with different exposure shares, net of anything common to all counties on a given day and anything fixed within a county. The date fixed effects absorb the national course of the campaign, including the response to the tape that was common to all counties; β is the additional change in the outcome per unit difference in the exposure share, and it is in these units that we report the estimates. Standard errors are clustered at the county level.

The event-study version replaces Post_t with indicators for 5-day bins relative to the

release,

$$y_{ct} = \sum_{k=[-40,-35]}^{[30,35]} \beta_k \text{Exposure Share}_c \times \mathbb{1}(t \in k)_t + \theta_c + \delta_t + \epsilon_{ct}, \quad (3)$$

where $\mathbb{1}(t \in k)$ indicates that date t falls in bin k ; the coefficients $\{\beta_k\}$ trace the response over time and test for pre-trends.

Permutation test. To complement the parametric standard errors, we use a permutation test for the difference-in-differences on the daily contribution panel: the exposure share is randomly reassigned across counties, one time-invariant value per county, while the outcome panel, the post indicator, and the fixed-effect structure are held fixed, and Equation (2) is re-estimated 1,000 times. The null is an arbitrary assignment of exposure across counties given the actual outcomes and the actual timing of the release, which is the relevant placebo for a design whose identifying variation is the within-county change around October 7. If the actual estimate lies in the tail of the permutation distribution, it is unlikely under this null.

Specificity. If the connection shaped what voters took in from the news, the response should be specific to the person the news concerned and to the connection itself. On the outcome side, we estimate Equation (2) for contributions to other Republican candidates and to Clinton with the same exposure share; a response that reflects the connection to Trump should not appear in giving to other candidates. On the treatment side, we use the Miss America counties of Section 3.6 in the placebo below and in the comparison group of Section 4.3.

Placebo: Miss America. A separate concern is whether the response reflects exposure to Trump specifically or some unobserved feature of pageant-producing communities. We address it with a placebo test that uses the Miss America counties of Section 3.6, which substitutes the treatment. We use the Miss America exposure share in place of the Miss USA share. If the response reflects a pageant-community channel rather than the connection to Trump, Miss America exposure should produce it too; if it reflects the connection, it should not.

4.2. Vote Intentions: Individual Exposure and Within-County Comparisons

Exposure. In parallel to the county share, we construct for each survey respondent a measure of her own exposure. Using the CCES, we count the Miss USA state titleholders from a respondent's county of residence who were crowned during Trump's ownership in years when the respondent was aged 10 or older:

$$\text{Exposure}_{icb} = \sum_{t=1996}^{2015} \mathbb{1}(t - b \geq 10)_{ibt} \times \text{Miss USA}_{ct}, \quad (4)$$

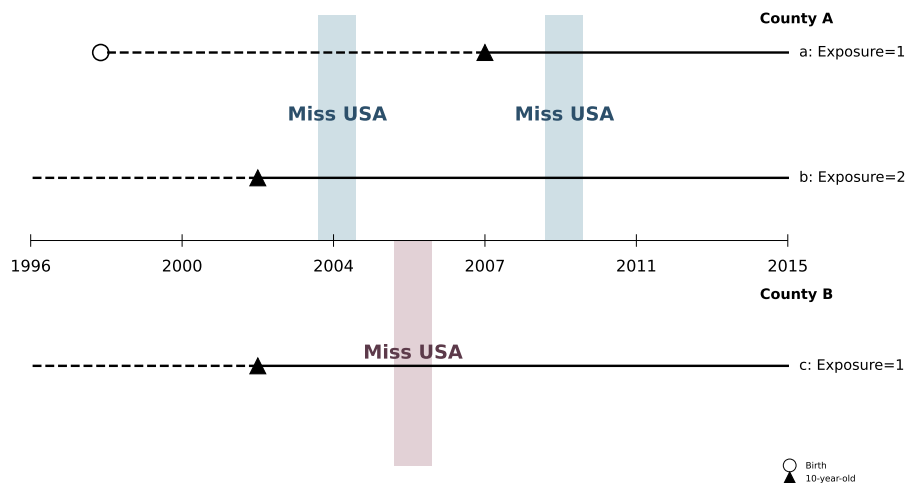


Figure 3. Research Design

Notes: Example of Individual-Level Exposure Design: A respondent born in 1990 in county A that produced Miss USA state titleholders in 2004 and 2009 would have an exposure measure of 2, as they were aged 14 and 19 during those years. In the same county, a respondent born in 1997 would have an exposure measure of 1, as they were aged 7 in 2004 (below the adolescence threshold) and 12 in 2009 (above the adolescence threshold). A respondent born in 1990 in county B that produced a Miss USA state titleholder in 2005 would have an exposure measure of 1.

where i indexes respondents, b is the respondent's birth year, and c is her county of residence. The indicator $\mathbb{1}(t - b \geq 10)_{ibt}$ equals one if the respondent was aged 10 or older in year t , and Miss USA_{ct} equals one if county c produced a Miss USA state titleholder that year. Summing their product over 1996–2015 gives the number of local titles the respondent was old enough to have followed during Trump's ownership. Figure 3 illustrates the construction.

Balance tests. The individual measure permits a comparison the county share cannot: respondents of different cohorts within the same county. If respondents with different exposure within a county already held systematically different political preferences, a differential change after the tape could reflect those differences rather than a response to the scandal. We therefore regress a set of pre-determined policy preferences—approval of the Garland Supreme Court nomination, support for repealing the Affordable Care Act, and attitudes toward Mexican border control—on exposure, conditioning on the full set of controls and fixed effects used below. None of the coefficients is statistically significant (Appendix Figure C4).

Specifications. The individual-level analysis is the same difference-in-differences design at a finer resolution. Because the CCES was fielded daily throughout the fall of 2016, some respondents were interviewed before October 7 and the rest after; combining

this pre–post structure with cross-cohort variation in exposure gives

$$Y_{icb\tau} = \alpha + \beta \text{Exposure}_{icb} \times \text{Post}_{\tau} + \rho \text{Exposure}_{icb} + \phi_b + \theta_{c\tau} + \Gamma' X_i + \epsilon_{icb\tau}, \quad (5)$$

where $Y_{icb\tau}$ is the outcome for individual i in county c and birth cohort b surveyed at date τ , and Post_{τ} equals one if the respondent was surveyed after October 7, 2016. Birth-cohort fixed effects ϕ_b absorb level differences across generations; county-by-survey-date fixed effects $\theta_{c\tau}$ absorb all time-varying county-level shocks—local economic conditions, campaign activity, media markets, and the local supply of news about the tape on that day—so that β is identified from respondents living in the same county, interviewed on the same day, who differ in how many titleholders they were old enough to have followed. This is the sharpest form of the comparison the paper makes: respondents who faced the same local information environment on the same day, and differed in the connection alone. X_i includes gender, education, race, employment status, party registration, social media usage, and ideological self-placement. Standard errors are clustered at the county level.

The event-study version replaces Post_{τ} with indicators for the day of interview relative to October 7:

$$Y_{icb\tau} = \alpha + \sum_{k=-3}^{5+} \beta_k \text{Exposure}_{icb} \times D_{k\tau} + \rho \text{Exposure}_{icb} + \phi_b + \theta_{c\tau} + \Gamma' X_i + \epsilon_{icb\tau}, \quad (6)$$

where $D_{k\tau}$ indicates being surveyed k days relative to the release; the $\{\beta_k\}$ trace the response day by day and test for pre-trends.

Permutation test. The permutation test of Section 4.1 has an individual-level counterpart: the exposure measure is randomly reassigned across respondents, with outcomes, interview dates, and fixed effects held fixed, and the specification is re-estimated 1,000 times.

4.3. Election Returns: Cross-Sectional Comparisons

The third outcome is the county vote share in the November election, to which the county exposure share of Section 4.1 applies. Vote shares are observed once per election and offer no timing variation within the campaign, so we estimate a cross-sectional regression:

$$Y_{cs} = \alpha + \beta \times \text{Exposure Share}_c + \phi_s + \Gamma' X_c + \epsilon_{cs}, \quad (7)$$

where Y_{cs} is the Republican presidential vote share in county c of state s , ϕ_s are state fixed effects, and X_c is a vector of county controls: the previous election's Republican vote share, poverty rate, median household income, employment, annual payroll,

number of firms, population, youth ratio, sex ratio, and Black, Hispanic, and Asian population shares. The exposure share for each election year is constructed as in Equation (1), using the eligible voting population in that year. Estimating Equation (7) for 2012 provides a pre-scandal placebo, and for 2020 and 2024 traces persistence. Standard errors are clustered at the state level. Because residents who moved in after a title are assigned the county’s history, Section 5.3 checks that the estimates are similar in counties with little in-migration, where the assignment is most accurate.

Randomization inference. One may worry that the timing of titles—whether a county produced a titleholder early or late in Trump’s ownership—is correlated with unobserved county characteristics that also shape its vote. We therefore use randomization inference in the spirit of Dell and Olken (2020). We re-simulate the pageant history 1,000 times, each time randomly reassigning, within each state-year, which county receives the titleholder among all counties that ever produced one; construct the exposure share implied by each simulated history; and re-estimate Equation (7). The procedure regenerates the timing of titles while preserving the structure of the pageant. If the actual estimate lies in the tail of the simulated distribution, it is unlikely under the reassignment null.⁹

Comparison group: Miss America counties. Miss America counties are pageant-producing communities without a connection to Trump: observationally equivalent to Miss USA counties (Section 3.6), but with a pageant he never owned. We re-estimate Equation (7) on the pooled sample of Miss USA and Miss America counties, assigning an exposure share of zero to counties that produced no Trump-era Miss USA titleholder, and on the propensity-score-matched sample of Section 3.6, in which each Miss America county is matched to a Miss USA county on observables. If the estimate reflects the connection to Trump rather than a feature of pageant communities, it should be similar across the baseline, pooled, and matched samples, and the Miss America exposure share should predict nothing.

5. The Response: Campaign Contributions, Vote Intentions, and Election Returns

This section documents the response in three steps. Section 5.1 compares counties: the daily contribution panel dates the response to the release and traces it across counties with different exposure shares. Section 5.2 compares individuals within counties: the daily survey contrasts cohorts in the same county interviewed on the same day, which

⁹This procedure differs from the permutation test above, which reassigns the exposure share itself given a fixed outcome panel; the two impose different nulls and are used where each is the relevant placebo.

rules out anything about high-exposure counties as such. Section 5.3 turns to election returns, which show whether the response reached the ballot box.

5.1. Campaign Contributions

Main estimates. Table 1 reports estimates of Equation (2). Because contributions in a county-day cell are highly skewed and nearly two-thirds of cells are zero, column (1) uses the inverse hyperbolic sine of contributions to Trump as the outcome.¹⁰ After October 7, contributions to Trump fall by more in counties with a higher exposure share. The coefficient on Exposure Share $\times$ Post is -0.54 IHS units per unit difference in exposure share. A one-standard-deviation difference in exposure share (0.47) therefore corresponds to a differential decline of about 0.25 IHS units after the release.¹¹ The exposure share is continuous, and we describe the estimates throughout as the differential associated with a unit difference in it.

Columns (2)–(4) express the same response in other units. In dollars, a unit difference in exposure share corresponds to about \$1,270 less per county-day given to Trump after the release, against a sample mean of \$842 over all county-days.¹² Among county-days with positive contributions, the log amount falls by 0.27 per unit of exposure share; and the probability that a county-day records any contribution to Trump at all falls by 3.8 percentage points, against a mean of 36 percent. The response therefore appears both in whether a county gives and in how much it gives; we return to this decomposition below.

Columns (5)–(7) establish that the response is specific to Trump. Contributions to other Republican candidates (column 5) and to Hillary Clinton (column 6) show no differential change with the exposure share: previously exposed communities did not turn against the party, nor did they redirect giving to the opponent.

Column (7) turns from the recipient of the contribution to the source of the connection. It reports the Miss America placebo of Section 4.1: in counties whose pageant connection is to Miss America rather than to Trump’s pageant, the Miss America exposure share is unrelated to contributions to Trump after the tape. A pageant tradition by

¹⁰The inverse hyperbolic sine, $\sinh^{-1}(y) = \log(y + \sqrt{y^2 + 1})$, behaves like $\log y$ for large y and is defined at zero, which makes it a common choice for skewed outcomes with many zeros (Burbidge et al., 1988; Bellemare and Wichman, 2020). Because its coefficients are not invariant to the units of y when zeros are present (Chen and Roth, 2024), we report the estimate in dollars, in logs on the positive cells, and on the extensive margin alongside it.

¹¹These estimates are changes in the transformed outcome, not percentage changes in contributions. If a percentage reading is nonetheless desired, the corresponding change is $\exp(-0.251) - 1 \approx -22$ percent, but with the large share of zero county-days this translation is itself only approximate.

¹²The dollar estimate is a level coefficient on a highly skewed outcome—nearly two-thirds of county-days record no contribution at all and the top one percent of county-days account for over a third of all dollars—so the sample mean is itself driven by a thin tail and the estimate is not best read as a share of it; against the standard deviation of county-day giving (about \$18,400), it is about seven percent.

Table 1. Campaign Contributions and the Access Hollywood Tape

	Contributions to Trump				Other recipients		Placebo	
	IHS(Y) (1)	Y (USD) (2)	log Y (3)	1(Y > 0) (4)	IHS, other Rep. (5)	IHS, Clinton (6)	Trump IHS, Miss America (7)	Trump IHS, Russiagate (8)
Exposure Share × Post	-0.536*** (0.084)	-1268.2*** (364.3)	-0.271*** (0.051)	-0.038*** (0.010)	0.073 (0.060)	-0.065 (0.043)		
Exposure Share (Miss America) × Post							-0.149 (0.111)	
Exposure Share × Post (Russiagate)								0.044 (0.044)
Date FE	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓
D.V. mean	2.676	842.4	6.678	0.363	3.483	4.695	1.399	3.360
N	49,197	49,197	17,839	49,197	49,197	49,197	30,084	49,197
R ²	0.548	0.031	0.555	0.473	0.616	0.817	0.374	0.621

Notes: The unit of observation is a county-day, September 1 to November 8, 2016. Exposure Share is the fraction of a county's voting-eligible residents exposed to a Trump-era Miss USA state titleholder (Equation 1); Post indicates dates after October 7, 2016. Columns (1)–(4) use contributions to Trump as the outcome: the inverse hyperbolic sine, the dollar amount, the log (county-days with positive contributions only), and an indicator for any contribution. Columns (5) and (6) use the inverse hyperbolic sine of contributions to other Republican candidates and to Hillary Clinton. Column (7) replaces the Miss USA exposure share with the Miss America exposure share, constructed by the same recipe, in the sample of Miss America counties that never produced a Trump-era Miss USA titleholder. Column (8) estimates the specification of column (1) over a different window, June 21 to August 28, 2016, and dates Post at July 27, 2016, the day Trump called on Russia to find the 30,000 emails missing from Hillary Clinton's private server; it is a placebo on the event date rather than on the treatment. All specifications include county and date fixed effects. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

itself did not produce the response; the connection to Trump did.

Column (8) turns from the treatment to the timing of the event. One might worry that residents of high-exposure counties simply pay more attention to news about Trump, whatever its content. We therefore re-estimate column (1) around a different Trump story, the one that broke on July 27, 2016, when he publicly called on Russia to find the emails missing from Hillary Clinton's private server.¹³ The story had the same candidate at its center and nothing to do with the pageant. We use a window of the same length as the baseline, June 21 to August 28, 2016, and date Post at July 27. The coefficient is small and insignificant: a Trump story by itself did not produce the response; the story that recalled the pageant did.

¹³"Russia, if you're listening, I hope you're able to find the 30,000 emails that are missing." The remark put Russian interference at the center of the campaign and made it the other running Trump story of the summer.

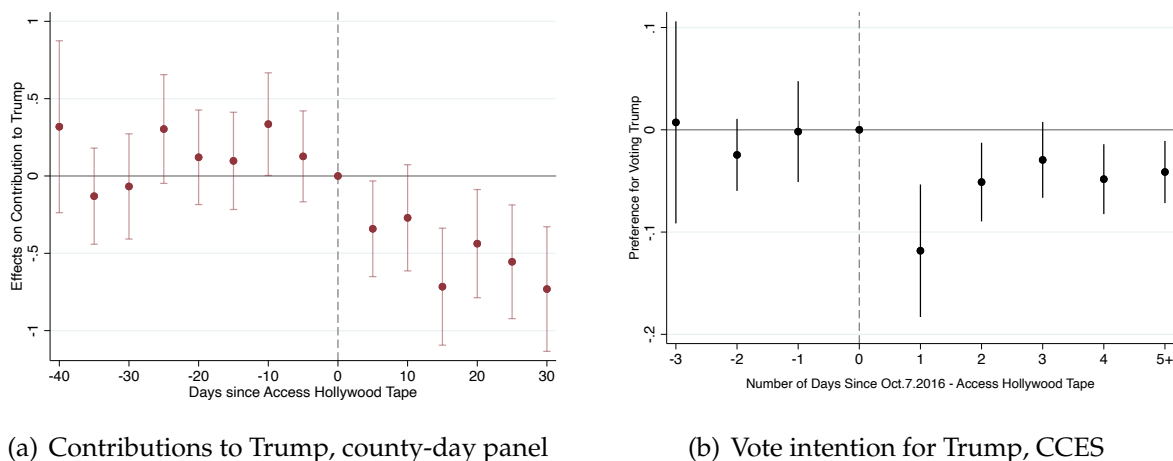


Figure 4. Event Studies: The Response Across Counties and Within Counties

Notes: Panel (a) plots the coefficients on the interaction between the county exposure share and indicators for 5-day bins relative to the release of the Access Hollywood tape, from Equation (3), with 99% confidence intervals; the outcome is the inverse hyperbolic sine of contributions to Trump, and standard errors are clustered at the county level. Panel (b) plots the coefficients on the interaction between the individual exposure measure and indicators for the day of interview relative to the release, from Equation (6); the outcome is the intention to vote for Trump, and the specification includes birth-year fixed effects, county-by-survey-date fixed effects, and the individual controls of Table 3. Standard errors are clustered at the county level.

Permutation test. Appendix Figure D1 reports the permutation test of Section 4.1 for the specification in column (1): we randomly reassign the county exposure share across counties, holding the contribution panel and the timing of the release fixed, and re-estimate the specification 1,000 times. The actual estimate lies far in the left tail of the resulting distribution ($p = 0.001$), so the estimate is unlikely to arise from a placebo assignment of exposure across counties.

Dynamics. Panel (a) of Figure 4 plots the event-study estimates of Equation (3) in 5-day bins. The coefficients before October 7 are small and statistically indistinguishable from zero across the 36-day pre-period, so counties with different exposure shares were on parallel paths before the release. After it, contributions to Trump fall by more where the exposure share is higher, and remain lower through election day.

Extensive and intensive margins. Table 2 decomposes the decline. After the release, a unit difference in exposure share corresponds to about two fewer contributions and 0.9 fewer distinct contributors per day, relative to means of 2.6 and 1.3—declines of roughly three-quarters of the mean. The amount per contribution and per contributor also fall, by about \$49 and \$73, but these are declines of about one-sixth of their means. The decline is therefore concentrated on the extensive margin of giving, in the number of contributions and of contributors, rather than in the amount per contribution. The

Table 2. Effects on Extensive Margin and Intensive Margin

	Contribution to Trump			
	# Donation (1)	# Donor (2)	USD per Donation (3)	USD per Donor (4)
Exposure Share $\times$ Post	-1.950*** (0.311)	-0.900*** (0.143)	-49.447** (24.870)	-72.877* (40.010)
Date FE	✓	✓	✓	✓
County FE	✓	✓	✓	✓
D.V. Mean	2.557	1.255	285.993	538.834
N	49197	49197	17839	17839
R ²	0.626	0.599	0.050	0.071

Notes: Exposure Share is the fraction of voting-eligible residents who were exposed to a Trump-era Miss USA state titleholder. The dependent variables are county date level number of donations to trump, number of donors who donated to Trump, average amount of USD per donation to trump, and average amount of USD per donor who donated to Trump. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

panel records county-day totals rather than individual donors, so it does not show whether the same people gave less or fewer people gave. Section 6.2 shows that the response is also stronger where the connection is more recent.

All U.S. counties. The baseline sample is restricted to pageant-producing counties because counties that never produced a titleholder differ from them in ways that could correlate with political outcomes (Section 3.6). Appendix Table D1 re-estimates columns (1)–(4) of Table 1 on all U.S. counties, assigning an exposure share of zero to counties that never produced a titleholder. The estimates are larger in every column and the larger estimates come from adding the comparison between pageant and non-pageant communities to the variation the baseline sample uses, and that comparison carries the selection the restriction was designed to remove; we therefore report them as a check that the response is not an artifact of the restriction.

5.2. Vote Intentions: The Response Within Counties

The county estimates compare places. A skeptic can grant them and still ask whether high-exposure counties differ in some way that made their residents turn against Trump when the tape arrived, independently of the pageant. The within-county design of Section 4.2 answers this.

Table 3 reports estimates of Equation (5) of Section 4.2, in which a respondent's exposure is the number of Trump-era titleholders from her county crowned when she was aged ten or older, and identification comes from comparing cohorts within a county

surveyed on the same day. Panel A uses the baseline sample of counties that have ever produced a titleholder; Panel B uses all counties. In column (1), a one-unit increase in exposure lowers the probability of intending to vote for Trump after October 7 by 2.9 percentage points in Panel A and 3.1 percentage points in Panel B, both significant at the 1% level, against a mean of about a third. The estimates are similar across the two samples and are consistent in direction with the county-level evidence.

The coefficient on Exposure itself, which measures the difference between more and less exposed cohorts before the tape, is positive in both panels (0.019 and 0.020). Cohorts with a stronger connection to the pageant did not hold a less favorable view of Trump before October 7; if anything, the reverse. The decline after the release is therefore not the continuation of an earlier aversion among these cohorts but a reversal of their relative position.

Panel (b) of Figure 4 plots the estimates of β_k from the event-study specification, Equation (6), day by day, beside the county-level event study of panel (a). There is no differential movement in the days before the release, supporting the parallel-trends assumption. After it, respondents with higher exposure show a decline in their intention to vote for Trump that peaks on the day following the release and remains negative through election day. The two panels show the same timing in two designs: a response that begins with the release and persists, across counties in giving and within counties in stated intentions.

Not before the tape. Column (3) uses the respondent’s primary vote for Trump, cast months before the release, as the outcome. Exposure bears no differential relationship to it after October 7, so the decline in column (1) is not a pre-existing difference between more and less exposed cohorts that happened to be surveyed later.

Robustness. The individual-level estimates survive the checks reported in the Appendix. Adding interactions of birth-year fixed effects, or of a linear birth-year trend, with survey-date fixed effects, which absorb any cohort-specific movement in attitudes over the campaign, leaves the estimate negative and significant (Appendix Table E3). A permutation test that reassigns exposure across respondents 1,000 times places the observed coefficient far outside the resulting distribution (Appendix Figure E1). Varying the age threshold from 6 to 12 in place of 10, the estimate is negative and significant for thresholds from 7 to 11 and loses significance at the extremes (Appendix Figure E2). The estimates are also robust for the exclusion of any single state (Appendix Figure E3).

Contributions and stated intentions both moved, in comparisons across counties and within them. Whether the vote did is the question of the next subsection.

Table 3. Access Hollywood Tape and the Shift of Attitudes

	Intends to vote for Trump	Trump's perceived conservatism	Voted for Trump in primary
	(1)	(2)	(3)
Panel A: Counties with Miss USA			
<i>Exposure × Post</i>	-0.029*** (0.010)	-0.103 (0.066)	-0.004 (0.012)
<i>Exposure</i>	0.019** (0.009)	0.098 (0.061)	0.001 (0.011)
D.V. Mean	0.318	5.144	0.123
N	16613	16146	20606
R ²	0.709	0.279	0.394
Panel B: Full Sample			
<i>Exposure × Post</i>	-0.031*** (0.010)	-0.102 (0.066)	-0.005 (0.012)
<i>Exposure</i>	0.020** (0.009)	0.088 (0.059)	0.001 (0.011)
Birth Year FE	✓	✓	✓
County × Survey Date FE	✓	✓	✓
Controls	✓	✓	✓
D.V. Mean	0.339	5.145	0.130
N	20700	20206	26032
R ²	0.723	0.298	0.417

Notes: This table presents estimates of the effect of pageant exposure on voting preferences and attitudes. The dependent variable in column (1) is a dummy indicating the immediate presidential vote preference for Trump. The dependent variable in column (2) is a 7-point assessment of Trump's ideological position. The dependent variable in column (3) is a dummy indicating whether the respondent voted for Trump in the primary election, which serves as a placebo. Panel A restricts the sample to counties that have ever produced a Miss USA state titleholder; Panel B includes the full sample. Controls include gender, education, race, employment status, party registration, social media usage, and ideological self-placement. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3. Election Returns

Two sources, with two different comparisons, show the same withdrawal of support: contributions across counties, dated to the release, and vote intentions within counties. This subsection asks whether it reached the ballot box. The vote is the consequential outcome, and neither source settles it: contributions could be confined to donors, and intentions are not votes. Election returns can offer complementary results to

Table 4. Election Returns: Vote Shares and Turnout

	Vote Share for				Turnout Rate
	Trump			Romney	
	2016 (1)	2020 (2)	2024 (3)	2012 (4)	2016 (5)
Exposure Share ²⁰¹⁶	-0.008*** (0.002)				-0.000 (0.003)
Exposure Share ²⁰²⁰		-0.005*** (0.002)			
Exposure Share ²⁰²⁴			-0.001 (0.001)		
Exposure Share ²⁰¹²				-0.001 (0.002)	
State FE	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓
D.V. Mean	0.556	0.566	0.586	0.543	0.444
N	707	707	707	707	707
R ²	0.969	0.986	0.996	0.986	0.740

Notes: The dependent variable is the Republican presidential vote share in the indicated election year. Exposure Share^{Year} is constructed according to the demographic structure of the eligible voting population in that year. Column (4) uses the 2012 Romney vote share as a pre-period placebo. Column (5) uses the 2016 turnout rate as the dependent variable. Controls include the previous election's Republican vote share, poverty rate, median household income, employment, annual payroll, number of firms, population, youth ratio, sex ratio, and Black, Asian, and Hispanic population shares. Standard errors are clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

alleviate this concern. Three checks guard against pre-existing differences across counties correlated with the exposure share: the pre-scandal placebo on the 2012 vote, randomization inference, and the comparison with Miss America counties, including on a propensity-score-matched sample.

Main estimate. We estimate Equation (7) of Section 4.3 for the 2016 election and, to trace persistence and to provide a pre-scandal placebo, for 2020, 2024, and 2012. Table 4 reports the results. Column (1) shows that in 2016 Trump's vote share was lower where the exposure share was higher: the coefficient of -0.008 , significant at the 1% level, implies a vote share 0.8 percentage points lower for a unit difference in exposure share.

Appendix Figure D2 applies the randomization-inference procedure of Section 4.3 to each election year: the 2016 estimate lies at the far left tail of the simulated distribution

($p < 0.001$) and the 2020 estimate is likewise in the left tail ($p = 0.01$), while the 2012 and 2024 estimates are unremarkable.

Later elections. Column (2) shows a negative association of 0.5 percentage points in 2020, significant at the 1% level, conditional on the previous election's Republican vote share. In 2024, the coefficient is small and statistically insignificant (column 3).

Pre-scandal placebo. Column (4) provides a placebo check using the 2012 Romney vote share, before the release of the tape. We find no significant effect, confirming that the negative association in 2016 is specific to the scandal rather than a pre-existing feature of high-exposure counties.

Turnout. To examine whether exposure is also associated with aggregate participation, we re-estimate Equation (7) with the county turnout rate as the dependent variable. Column (5) of Table 4 shows no statistically significant relationship between exposure and 2016 turnout. We therefore find no evidence of a corresponding change in aggregate turnout.

Miss America placebo and comparison group. We apply the comparison-group design of Section 4.3 and the placebo of Section 4.1 to election returns. Appendix Table D2 reports the results across all four election years. Columns (1)–(4) present estimates in the full pooled sample; columns (5)–(8) present estimates in the propensity-score-matched sample. The 2016 coefficient is -0.008 in the pooled sample, significant at the 1% level, and -0.004 in the matched sample, significant at the 5% level. The 2020 coefficient is also negative, at -0.004 , and significant at the 5% level in both samples, conditional on the 2016 Republican vote share. The coefficients for 2012 and 2024 are small and statistically insignificant in both samples. The similarity of the pooled and matched estimates indicates that Miss America counties are already a well-matched comparison group and that the result is not sensitive to tightening the comparison. The placebo that substitutes the Miss America exposure share for the Miss USA share, applied to vote shares, likewise shows no effect (Appendix Table D3).

Robustness. The 2016 estimate is robust to how exposure is measured and to the sample. Three alternative exposure measures—an intensity-weighted share that counts the number of Trump-era titleholders each cohort was old enough to follow, an indicator for counties whose pageant activity was disproportionately concentrated in the Trump era, and the raw Trump-era count entered jointly with the all-period count—give negative and statistically significant estimates (Appendix Table D4). Appendix Table D5 varies the sample. Excluding counties in the tails of the population distribution, below the 5th or 10th percentile or above the 90th or 95th, leaves the estimate negative and significant (columns 1–4). Because residents who moved in after a title was won are

assigned the county's full exposure history, in-migration introduces measurement error; restricting the sample to counties with low mobility rates in the American Community Survey, under 3, 5, or 10 percent, yields estimates of similar magnitude and significance (columns 5–7).

6. Inside the Response: Attention, Memory, and Updating

Section 5 showed that voters with a prior connection to the pageant withdrew support from Trump when the tape was released. This section asks how. A connection formed years earlier, outside politics, can shape the response to news only through the voter herself: through whether she acquires the news and attends to it, through what the news brings to mind, and through what she revises. Sections 6.1, 6.2, and 6.3 take these in turn, drawing on search activity, which records whether previously exposed communities sought out the news, and on the daily survey, which records what individual voters revised and who among them did so. We read the evidence as patterns consistent with each mechanism rather than as further causal estimates; Section 7 then organizes the three in a framework.

6.1. Acquisition and Attention

Did previously exposed voters attend more to the tape? The first evidence comes from search activity, using the county-week panel of Section 3.5. We estimate Equation (2) at the weekly frequency, with county and week fixed effects, so that the comparison is across counties within a week and within a county across weeks.

Table 5 reports the results. After the release, search interest in Trump rises by more where the exposure share is higher (column 1), as a story about him should produce, and so does search interest in the tape itself (column 3), with a coefficient significant at the 1 percent level and large relative to the mean. In the Miss America counties that never produced a Trump-era Miss USA titleholder, the Miss America exposure share predicts neither (columns 2 and 4). The rise in attention reflects the connection to Trump's pageant, not a pageant tradition.

Columns (5) and (6) are the search counterpart of the timing placebo in Section 5.1. If residents of high-exposure counties simply followed Trump more closely, their attention would rise with any story about him. Russian interference, the other Trump story of the campaign, ran through the same weeks and had nothing to do with the pageant; neither the exposure share nor the Miss America share predicts search interest in it. As with contributions, the connection directed attention to the story that recalled the pageant, not to Trump news in general.

Table 5. Attention to the Scandal

	Google Trends Search Interest					
	'Trump'		'Access Hollywood'		'Russian Interference'	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure Share $\times$ Post	0.953*** (0.146)		0.848*** (0.172)		0.089 (0.057)	
Exposure Share (Miss America) $\times$ Post		-0.120 (0.234)		0.120 (0.106)		0.026 (0.030)
Week FE	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓
D.V. Mean	8.548	5.044	0.616	0.189	0.045	0.016
N	7000	4290	7000	4290	7000	4290
R ²	0.965	0.937	0.345	0.155	0.100	0.103

Notes: Exposure Share is the fraction of voting-eligible residents who were exposed to a Trump-era Miss USA state titleholder. Exposure Share (Miss America) is the fraction of voting-eligible residents who were exposed to a Trump-era Miss America state titleholder, in the sample of Miss America counties that never produced a Trump-era Miss USA titleholder. The unit of observation is a county-week, September 1 to November 8, 2016; Post indicates weeks after October 7, 2016. The sample is the baseline counties whose DMA Google Trends reports (columns 1, 3, 5) and the Miss America counties that never produced a Trump-era Miss USA titleholder (columns 2, 4, 6). The dependent variables are Google Trends search interest for "Trump," "Access Hollywood," and "Russian interference," assigned to counties from their DMA in proportion to population. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The CCES gives a second look from the individual side which suggests the attention channel. The survey asks whether the respondent had recently read a story or watched a video about politics, which we take as an indicator of being in the audience for political news. Exposure does not predict this indicator after the release, so the connection did not draw in people who were not already following politics; and the decline in support for Trump is concentrated among those who were, and indistinguishable from zero among those who were not (Appendix Table E4).

Taken together, search interest in the tape and in the candidate it concerned rose after the release where the connection was stronger, interest in the other Trump story did not, and the withdrawal of support appears only among voters who were consuming political news. If this attention came from voters for whom the tape brought a local title to mind, it should fade as that memory does. The next subsection looks for this decay.

6.2. Memory and Recency

If the connection operates through memory, a connection formed more recently should be recalled more readily when the cue arrives and should produce a stronger response. We examine this at both levels in Table 6. Panel A re-estimates the contribution specification, Equation (2), on subsamples that the exposed counties whose most recent

Trump-era titleholder was crowned fewer than 5, 10, 15, and 20 years before the election and never-exposed counties (columns (1)–(4)). The estimate rises monotonically in magnitude as the window narrows, from -0.54 in the 20-year window, which contains nearly the full sample, to -0.66 in the 5-year window, a difference of about a fifth. Each estimate is significant at the 1% level, and the mean of the outcome is nearly identical across columns, so the pattern does not reflect differences in the level of giving across subsamples.

Panel B repeats the exercise for individual vote intentions, keeping respondents by the recency of their most recent exposure. The pattern is the same. The decline in support for Trump per unit of exposure is 2.9 percentage points among respondents whose most recent exposure was within twenty years and 3.6 percentage points among those exposed within five years, each significant at the 1% level. Two outcomes, at two levels of aggregation, thus show the point estimates growing with the recency of the connection. Because the subsamples are nested, the differences between adjacent windows are within sampling error; we read the gradient as consistent with the decay of recall with time, to which Section 7 relates it, rather than as a test of it. Attention and memory describe how the news was taken in. What did voters do with it?

6.3. Updating: Support, Not Positions

Support, not ideological placement. Column (2) of Table 3 in Section 5.2 asks whether the tape changed where respondents placed Trump on the ideological spectrum. It did not: exposure has no differential effect on his perceived conservatism after the release in either panel. The contrast—lower support with no change in ideological placement—is consistent with previously exposed voters reassessing the candidate rather than his positions.

Trump, not the party. Appendix Table E1 replaces the outcome with vote intentions for Republican candidates for the House, the Senate, and governorships. The coefficient on Exposure $\times$ Post is small and insignificant for all three. The response stayed with the person the tape concerned, as it did in the contribution data.

Not identity. Appendix Table E2 adds to Equation (5) triple interactions of Exposure $\times$ Post with indicators for female, white, college-educated, employed full-time, and aged 60 or older. None is significant, and the main coefficient is unchanged throughout. The response is not concentrated in any demographic group; in particular, women did not respond more than men, despite the content of the tape, which weighs against a reading in which the response expresses group identity rather than a revised assessment of the candidate.

Together, these patterns describe a voter who attended to the news because it

Table 6. Recency of Exposure

Years since most recent exposure	< 5 (1)	< 10 (2)	< 15 (3)	< 20 (4)
Panel A: Contributions to Trump (county-day, IHS)				
Exposure Share $\times$ Post	−0.656*** (0.107)	−0.585*** (0.093)	−0.553*** (0.087)	−0.535*** (0.084)
Date FE	✓	✓	✓	✓
County FE	✓	✓	✓	✓
D.V. mean	2.602	2.704	2.691	2.664
N	28,773	36,639	42,780	48,369
R ²	0.550	0.554	0.548	0.546
Panel B: Intends to vote for Trump (respondent)				
Exposure $\times$ Post	−0.036*** (0.011)	−0.029*** (0.010)	−0.029*** (0.010)	−0.029*** (0.010)
Birth year FE	✓	✓	✓	✓
County $\times$ survey date FE	✓	✓	✓	✓
Controls	✓	✓	✓	✓
D.V. mean	0.288	0.284	0.293	0.304
N	6,565	10,139	11,638	13,102
R ²	0.688	0.685	0.689	0.695

Notes: Each column restricts the sample by the recency of the most recent Trump-era Miss USA state titleholder, with the window in years given in the column header. Panel A estimates Equation (2) with county and date fixed effects on the county-day panel of contributions to Trump, in inverse hyperbolic sine; each subsample keeps the counties whose most recent Trump-era titleholder was crowned fewer than the stated number of years before the election, together with counties never exposed during Trump’s ownership. Standard errors are clustered at the county level. Panel B estimates Equation (5) with birth-year fixed effects, county-by-survey-date fixed effects, and individual controls on CCES respondents whose most recent exposure occurred fewer than the stated number of years before the election. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

touched something she remembered, and who revised her assessment of the person rather than of his positions. Section 7 gives this chain a formal statement.

7. A Model of News Filtering

The evidence of Sections 5 and 6 can be organized by a simple model of news filtering. An event carries a cue; the cue retrieves whatever in the voter’s past resembles it; what is retrieved lowers the cost of attention; and attention drives updating. This section sets out a framework that makes each link explicit, derives four predictions from it, and then returns to the scandal we study to show how it enters the framework and how each prediction corresponds to a finding.

7.1. Model and analysis

Beliefs. A voter i wants to know the candidate’s fitness for office, $\theta \in \mathbb{R}$ —a scalar summary of his caliber and character—with higher θ denoting a fitter candidate. Her prior is $\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$, with precision $h_0 \equiv 1/\sigma_0^2$. Her objective is accuracy: she minimizes the expected squared error of her belief, $\mathbb{E}[(\hat{\theta}_i - \theta)^2]$, net of the cost of learning.

The event and its cue. An event n occurs. Its *cue* is a set of visible features $x_n \subseteq F$ —the actors named, the institution involved, the kind of story—that is available to every voter as soon as the event occurs. The event is informative about θ , but the voter learns from it only to the extent that she attends to it.

Memory and connection. The voter’s past experience is a database of episodes, $\mathcal{D}_i = \{(e_k, t_k)\}_{k=1}^{K_i}$, each a set of features $e_k \subseteq F$ encoded at date t_k . A cue brings episodes to mind in proportion to their similarity to it, $S(x_n, e_k, t_k) = |x_n \cap e_k| g(t - t_k)$, the number of shared features discounted by a forgetting function g with $g(0) = 1$ and $g' \leq 0$. This is the feature-based similarity of Tversky (1977), as used in the memory models of Mullainathan (2002) and Bordalo et al. (2020, 2023). Recall is competitive: episodes compete for a limited capacity to come to mind. The *connection* between the voter and the event is the share of recall that the cue captures,

$$c_i(n) = \frac{A_i(n)}{A_i(n) + \rho K_i}, \quad A_i(n) = \sum_{k=1}^{K_i} S(x_n, e_k, t_k), \quad (8)$$

where $A_i(n)$ is the activation the cue produces in memory and ρK_i is the interference from everything else the voter remembers. The ratio is the competitive retrieval rule of associative-memory models (Kahana, 2012; Bordalo et al., 2023), in which activation by the cue competes with interference from the rest of memory; the forgetting function captures recency (Gennaioli et al., 2024). Connection is zero if nothing in memory shares a feature with the cue, and is higher the more related episodes the voter holds, the more features they share with the cue, and the more recent they are. Connection is not chosen: the cue is set by the event and the memory was formed before it.

Attention. By paying attention $a \geq 0$ to the event, the voter acquires information about it, summarized by a signal $s = \theta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2/a)$: more attention, less noise; $a = 0$ means the voter ignores the event. This is the attention technology of Myatt and Wallace (2012), in which the noise a receiver adds to a source falls in proportion to the attention she pays it. Attention costs $\psi_i(n) a$, a constant marginal cost of precision

in the sense of Pomatto et al. (2023), with

$$\psi_i(n) = \frac{\psi}{1 + \phi c_i(n)}, \quad \psi, \phi > 0, \quad (9)$$

so that an event one has a connection to is cheaper to attend to—easier to absorb, or less onerous to follow, in the spirit of information costs that depend on the similarity between what is to be learned and what is already known (Hébert and Woodford, 2021).¹⁴

The attention decision. Given the cue, the voter chooses attention and a rule for forming her belief from the signal to minimize expected squared error plus the cost of attention:

$$\min_{a \geq 0, \hat{\theta}(\cdot)} \mathbb{E}[(\hat{\theta}(s) - \theta)^2] + \psi_i(n) a. \quad (10)$$

The voter decides whether and how much attention to pay to the event. For any $a > 0$ the error-minimizing rule is the posterior mean and the minimized error is the posterior variance, $1/(h_0 + a/\sigma^2)$, so the choice of attention reduces to $\min_{a \geq 0} 1/(h_0 + a/\sigma^2) + \psi_i(n) a$. The objective is convex. Its first-order condition equates the marginal reduction in error, $\sigma^{-2}(h_0 + a/\sigma^2)^{-2}$, to the marginal cost $\psi_i(n)$; the corner $a = 0$ obtains when the marginal reduction in error at zero attention, σ_0^4/σ^2 , already falls short of the cost, that is, when the prior is precise enough relative to the cost and the noise that learning is not worthwhile. The solution is

$$a_i^*(n) = \max \left\{ 0, \frac{\sigma}{\sqrt{\psi_i(n)}} - \sigma^2 h_0 \right\}. \quad (11)$$

Her belief is then

$$\hat{\theta}_i = \begin{cases} \mu_0 + \kappa_i (s - \mu_0), & \kappa_i = 1 - \frac{\sigma \sqrt{\psi_i(n)}}{\sigma_0^2}, \quad \text{if } a_i^*(n) > 0, \\ \mu_0, & \text{if } a_i^*(n) = 0, \end{cases} \quad (12)$$

where κ_i is the weight she places on what she learns. Given the truth, her expected revision is $\mathbb{E}[\hat{\theta}_i - \mu_0 \mid \theta] = \kappa_i(\theta - \mu_0)$ if she attends and zero otherwise. An unfavorable event—one revealing the candidate to be less fit than the prior held, $\theta < \mu_0$ —lowers the voter’s assessment of him, and by more the larger the gain.

Predictions The connection c is fixed for the voter once the event occurs: the cue is set by the event and the memory was formed before it, so c varies across voters through

¹⁴The idea that what is already known lowers the cost of absorbing what is new has a long history, from absorptive capacity (Cohen and Levinthal, 1990) to processing fluency in psychology (Reber et al., 2004).

what they remember and how long ago it was formed. Writing the cost of attention as $\psi(c) = \psi / (1 + \phi c)$, attention and the gain become functions of c , and the predictions of the framework are comparative statics in c , and in the age of the experience behind it.

Proposition 1. *Fix the event, the prior (μ_0, σ_0^2) , and the noise σ^2 .*

- (i) *Acquisition margin. The voter attends to the event if and only if $c > \bar{c} \equiv \frac{1}{\phi} (\frac{\psi \sigma^2}{\sigma_0^4} - 1)$.*
- (ii) *Attention level. Conditional on attending, $a^*(c) = \sigma \sqrt{(1 + \phi c) / \psi} - \sigma^2 h_0$ is strictly increasing in c .*
- (iii) *Updating. For an unfavorable event, $\theta < \mu_0$,*

$$\frac{\partial}{\partial c} \mathbb{E}[\hat{\theta} - \mu_0 \mid \theta] = \kappa'(c) (\theta - \mu_0) < 0 : \quad (13)$$

the voter's assessment of the candidate falls, and falls by more the stronger her connection to the event.

- (iv) *Recency of the experience. Let the voter's memory contain one episode related to the cue, encoded $\tau = t - t_k$ periods before the event. Then c is decreasing in τ , and so for an unfavorable event the fall in the voter's assessment is smaller the older the experience,*

$$\frac{\partial}{\partial \tau} \mathbb{E}[\hat{\theta} - \mu_0 \mid \theta] = \kappa'(c) \frac{\partial c}{\partial \tau} (\theta - \mu_0) \geq 0, \quad (14)$$

with strict inequality where $g' < 0$. An experience old enough that c falls below $\bar{c}$ produces no response at all.

Derivations of (11) and (12) and the proof of the proposition are in Appendix G.

7.2. The Scandal in the Framework

The scandal is an event interpreted through the framework: its cue, as it reached voters through the coverage, includes the features {Trump, Miss USA, pageant}, and what the tape revealed about the candidate is the information that required attention. A voter who lived in a county that produced a titleholder during Trump's ownership, and was old enough to follow it, is likely to hold an episode $e^P = \{\text{Trump, Miss USA, pageant, hometown}\}$ that shares features with the cue; by (8) she has a larger connection c to the scandal than a voter with no such episode, and a larger one the more recent the title.

- (i) *Acquisition margin.* A connection can move a voter from ignoring an event to attending to it, so where this margin is active it changes how many people respond rather than how much each responds. This is related to the contribution response,

which is concentrated in the number of contributions and of contributors; the pattern is consistent with part (i).

(ii) *Attention level.* Among those who attend, a stronger connection raises attention to the event, and because the cost reduction in (9) is specific to events whose cue recalls something the voter holds, it raises attention to this event without raising attention to news in general. That is the pairing in Section 6.1: search for the scandal rises with the exposure share while political news consumption does not.

(iii) *Updating.* Part (iii) concerns what happens once a voter attends. Her revision is $\kappa(c)(\theta - \mu_0)$: the event sets its sign, the connection its size. The tape was unfavorable news, $\theta < \mu_0$ (Section 2.3), so the model predicts that assessments of Trump fell, and fell by more where the connection was stronger. The exposure measures of Section 4 are proxies for c —the county share for its average level in the electorate, the individual count for its level for a respondent—so the coefficient on Exposure $\times$ Post estimates the differential in (13), up to the scaling of exposure into c and of assessment into support. Its sign and its ordering across exposure are the prediction, and both hold in contributions, vote intentions, and vote shares alike.

(iv) *Recency.* The connection, and with it the response, decays with the age of the experience through the forgetting function g in (8), so the same event moves a voter less the longer ago the experience was formed. This is the recency gradient of Section 6.2, at both levels, as (14) predicts.

Two features bear on interpretation. Connection enters only through the cost of attention, so (13) says nothing about the level of support before the event: the design identifies a differential response to the scandal, not a pre-existing premium or penalty. And the differential arises because the cue reactivates a specific stored episode, not because the candidate is familiar in general.

8. Conclusion

This paper asks whether voters respond more strongly to a scandal about a politician when it concerns something to which they already have a personal or local connection. The release of the Access Hollywood tape provided news that was available to every voter at once; Trump’s two decades as owner of the Miss USA pageant provided a connection, formed years earlier and outside politics, that can be traced to a county and a birth year. Where the connection was stronger, contributions to Trump fell at once and stayed lower through election day, and his vote share was lower in November and again four years later. The response was timed to the tape and specific to Trump, larger in the point estimates where the connection was more recent, and concentrated among voters who followed political news.

The finding bears on how information produces accountability. Studies of disclosure vary what voters are told and find that they punish what they learn; the disclosure here was the same for everyone, and the response still differed with what voters brought to it. Reception is thus a margin of its own, and it has a geography. A politician's past—a business, an institution, a hometown—leaves connections distributed unevenly across the electorate, and news about that past lands hardest where it is remembered. The same disclosure can produce accountability in some communities and none in others, for reasons that have nothing to do with how the news was supplied or with the partisan lean of those who received it. Which communities carry a connection to a politician's past is therefore a determinant of accountability in its own right.

The mechanism we propose is not confined to politics. It predicts that a corporate scandal is taken in most by consumers and investors with a prior tie to the firm formed outside the scandal, such as a hometown, a former employer, or a familiar product; that news about a disease reaches those who have seen it in someone they know; and that a distant disaster moves those with a connection to the place. Evidence from these domains would be a natural next step.

Two questions remain open. Whether a connection changes the content of what voters take in, or only the amount, is beyond what our evidence can establish. And whether the mechanism operates for favorable news as well—where the framework implies the mirror image, a rise in support among the previously exposed—is a question only a different event can answer. We leave both to future work.

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Online Appendix

(Not intended for publication)

A. Background Evidence

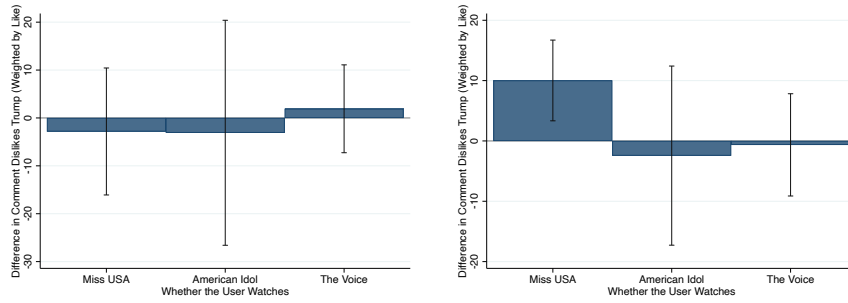


Figure A1. YouTube Comments on the Presidential Debate

Notes: This figure displays sentiment toward Trump in YouTube comments on the 2016 presidential debate livestreams before and after the Access Hollywood tape, separately by program affiliation (Miss USA, American Idol, The Voice). Negative-sentiment indices are weighted by comment likes.

Table B1. County-Level Summary Statistics

	Obs.	Mean	S.D.
Republican vote share 2016	708	0.555	0.165
ExposureShare2016	708	0.571	0.469
ExposureShare (Miss America)	448	0.596	0.468
Republican vote share 2012	708	0.542	0.150
Republican vote share 2008	708	0.520	0.137
Republican vote share 2004	708	0.565	0.127
Republican vote share 2000	708	0.535	0.121
Poverty rate	708	14.359	5.351
Median household income (1k)	708	55.296	15.614
Employment (1k)	708	123.826	277.156
Annual payroll (millions)	708	6.683	18.065
Number of firms (1k)	708	6.385	14.175
Population (millions)	708	0.297	0.630
Youth ratio	708	0.100	0.040
Sex ratio	708	0.983	0.071
Black proportion	708	0.103	0.136
Asian proportion	708	0.027	0.043
Hispanic proportion	708	0.093	0.118

Notes: This table reports summary statistics for variables used in the county-level analysis.

B. Data and Sample

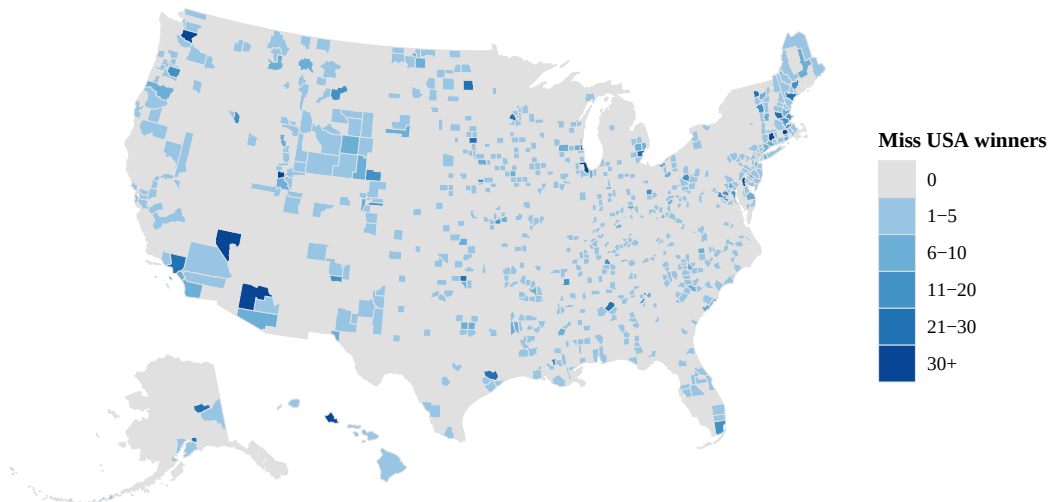


Figure B1. Geographic Distribution of Miss USA State Titleholders

Notes: This figure illustrates the geographic distribution of Miss USA state titleholders across U.S. counties from 1952 to 2015.

Table B2. Individual-Level Summary Statistics

	Obs.	Mean	S.D.
Preference for Voting Trump	40,573	0.413	0.492
Exposure	40,573	1.534	2.823
Birth year	40,573	1,965.637	16.567
Male	40,573	0.480	0.500
Education: No HS	40,573	0.019	0.136
Education: High school graduate	40,573	0.224	0.417
Education: Some college	40,573	0.243	0.429
Education: 2-year	40,573	0.114	0.317
Education: 4-year	40,573	0.253	0.435
Education: Post-grad	40,573	0.148	0.355
Race: White	40,573	0.729	0.444
Race: Black	40,573	0.131	0.337
Race: Hispanic	40,573	0.074	0.262
Race: Asian	40,573	0.026	0.161
Race: Native American	40,573	0.008	0.086
Race: Mixed	40,573	0.019	0.135
Race: Other	40,573	0.011	0.107
Race: Middle Eastern	40,573	0.002	0.040
Employment Status: Full-time	40,573	0.450	0.497
Employment Status: Part-time	40,573	0.102	0.302
Employment Status: Temporarily laid off	40,573	0.006	0.075
Employment Status: Unemployed	40,573	0.042	0.200
Employment Status: Retired	40,573	0.236	0.425
Employment Status: Permanently disabled	40,573	0.055	0.229
Employment Status: Homemaker	40,573	0.058	0.234
Employment Status: Student	40,573	0.034	0.181
Employment Status: Other	40,573	0.018	0.132
Party Registration: No Party, Independent, Decline to state	40,573	0.218	0.413
Party Registration: Democratic Party	40,573	0.470	0.499
Party Registration: Republican Party	40,573	0.293	0.455
Party Registration: Other	40,573	0.008	0.088
Social Media Forwarded Politics	40,573	0.234	0.423
Ideology Score(5)	40,556	3.075	1.216

Notes: This table reports summary statistics for variables used in the CCES analysis.

Table B3. Miss USA Counties Compared with Miss America Counties and with All Counties

	Dep. Var.: 1(Counties with Miss USA)	
	Counties with Miss USA or Miss America (1)	All Counties (2)
Republican vote share 2012	-0.143 (0.165)	0.066 (0.080)
Poverty rate	-0.008 (0.006)	-0.006** (0.002)
Median household income (1k)	0.000 (0.002)	0.005*** (0.001)
Employment (1k)	0.000 (0.001)	0.003*** (0.000)
Annual payroll (millions)	-0.000 (0.005)	-0.024*** (0.003)
Number of firms (1k)	-0.003 (0.006)	-0.008* (0.004)
Population (millions)	0.085 (0.150)	-0.188* (0.108)
Youth ratio	0.158 (0.360)	1.744*** (0.195)
Sex ratio	0.170 (0.208)	-0.250*** (0.063)
Black proportion	0.198 (0.205)	0.399*** (0.092)
Asian proportion	0.027 (0.613)	3.151*** (0.402)
Hispanic proportion	0.113 (0.175)	0.323*** (0.079)
State FE	✓	✓
D.V. Mean	0.446	0.325
N	1588	3558
R ²	0.053	0.326

Notes: This tables compares the characteristics of counties with Miss USA state titleholders to those with Miss America state titleholders and all counties. The dependent variable is an indicator for whether a county has a Miss USA state titleholder. The first column restricts the sample to counties with Miss USA state titleholders and counties with Miss America state titleholders, while the second column includes all counties. All regressions include state fixed effects. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C. Balance Tests

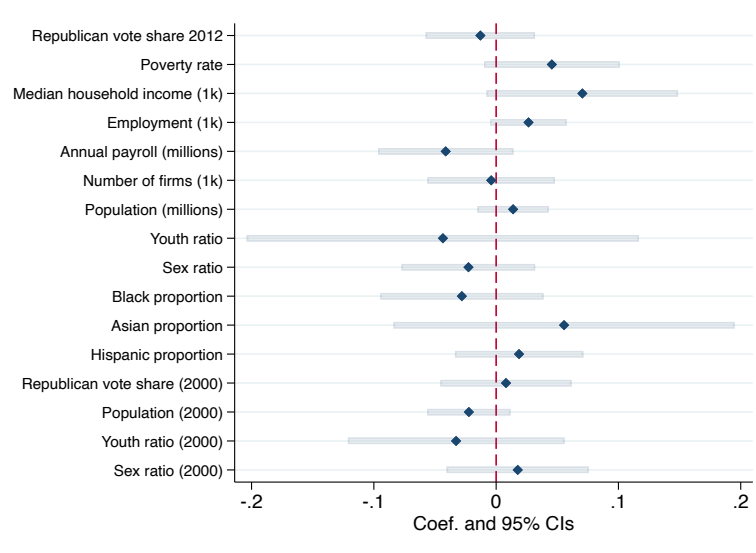


Figure C1. County Exposure Balance Test

Notes: This figure plots the coefficients of regressing a set of county-level characteristics on the county-level exposure share, conditioning on state fixed effects and additional covariates.

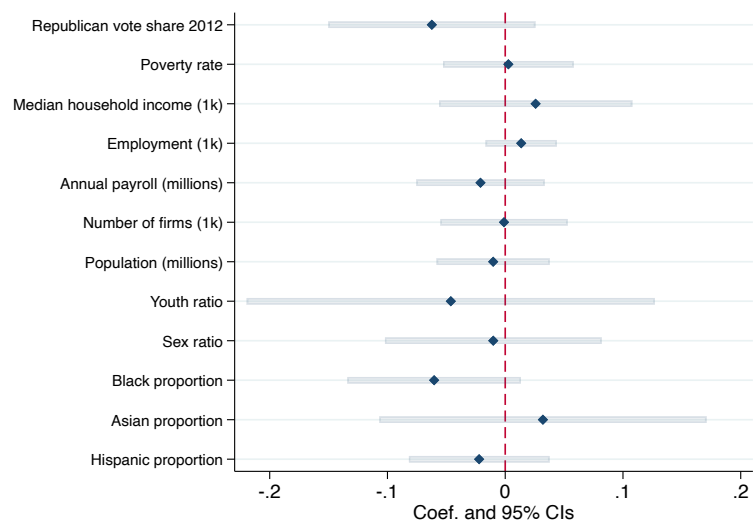


Figure C2. Timing of the Winning Balance Test

Notes: This figure plots the coefficients of regressing a set of county-level characteristics on the EarlyWinner, a dummy indicating the county has at least one Miss USA state titleholder in 1996–2005, conditioning on state fixed effects, the number of Miss USA state titleholders, and additional covariates.

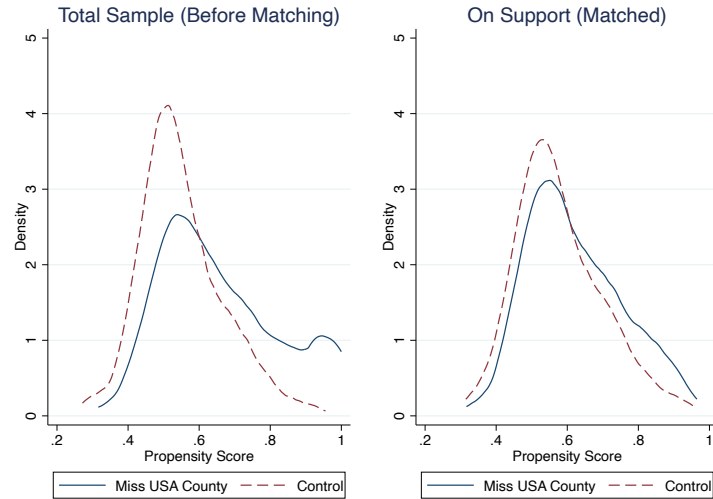


Figure C3. Propensity Score Distribution

Notes: This figure plots the propensity score distribution for Miss USA and Miss America counties.

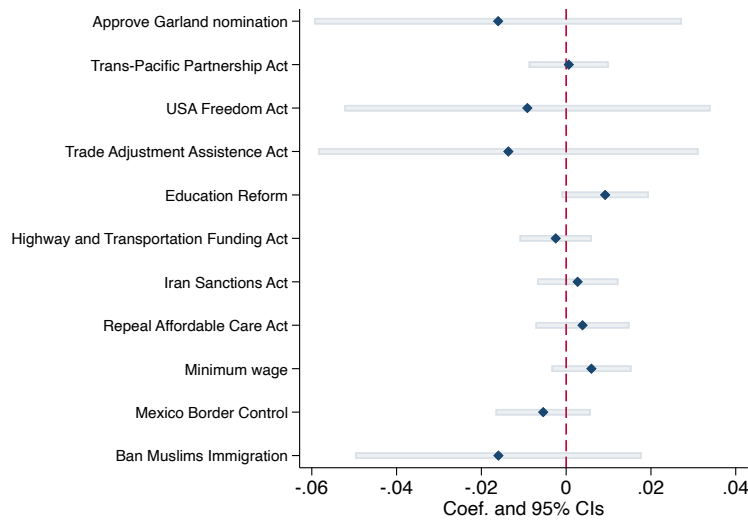


Figure C4. Individual Exposure Balance Test

Notes: This figure plots the coefficients of regressing a set of pre-determined policy preferences on the individual-level exposure measure, conditioning on individual characteristics, birth year fixed effects, and county-by-survey-date fixed effects. Robust standard errors are clustered at the county level.

D. Additional County-Level Results

Table D1. Campaign Contributions and the Access Hollywood Tape: All U.S. Counties

	Contributions to Trump			
	IHS(Y) (1)	Y (USD) (2)	log Y (3)	$\mathbb{1}(Y > 0)$ (4)
Exposure Share $\times$ Post	-1.036*** (0.057)	-1521.3*** (341.4)	-0.357*** (0.034)	-0.093*** (0.007)
Date FE	✓	✓	✓	✓
County FE	✓	✓	✓	✓
D.V. Mean	1.189	266.6	6.399	0.168
N	222,180	222,180	36,966	222,180
R ²	0.456	0.031	0.508	0.401

Notes: This table re-estimates columns (1)–(4) of Table 1 on all U.S. counties rather than the baseline sample of pageant-producing counties. Counties that never produced a Miss USA state titleholder are assigned an exposure share of zero. The unit of observation is a county-day, September 1 to November 8, 2016. Exposure Share is the fraction of a county’s voting-eligible residents exposed to a Trump-era Miss USA state titleholder (Equation 1); Post indicates dates after October 7, 2016. The outcomes are contributions to Trump: the inverse hyperbolic sine, the dollar amount, the log (county-days with positive contributions only), and an indicator for any contribution. All specifications include county and date fixed effects. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

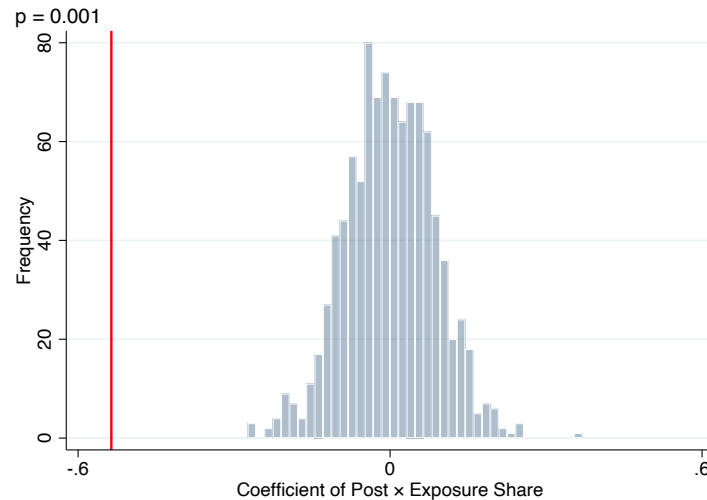


Figure D1. Permutation Test: Campaign Contributions

Notes: This figure plots the distribution of the coefficient on Exposure Share $\times$ Post from Equation (2) across 1,000 permutations in which the county exposure share is randomly reassigned across counties, one value per county, with the contribution panel, the post indicator, and the fixed effects held fixed. The vertical line marks the actual estimate.

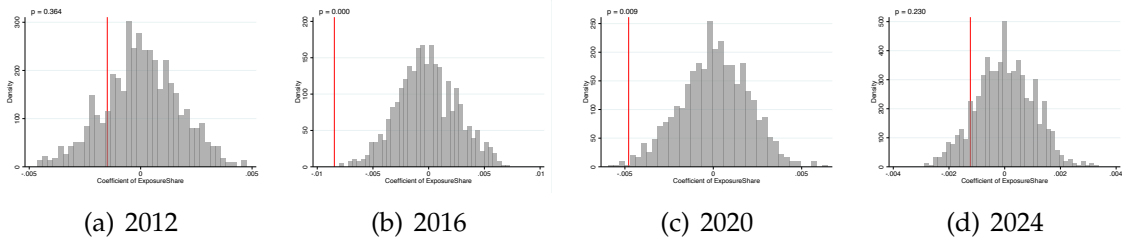


Figure D2. Randomization Inference: Election Returns

Notes: This figure plots the distribution of the estimated coefficients from the randomization inference, where the county-level exposure share is re-simulated 1,000 times by randomly assigning the Miss USA state titleholders across counties within each state-year. The vertical line indicates the observed coefficient from the main specification.

Table D2. Election Returns: Pooled and Propensity-Score-Matched Samples

	Republican Vote Share							
	Pooled sample				Matched sample			
	2012 (1)	2016 (2)	2020 (3)	2024 (4)	2012 (5)	2016 (6)	2020 (7)	2024 (8)
Exposure Share ²⁰¹²	-0.002 (0.002)				-0.003 (0.002)			
Exposure Share ²⁰¹⁶		-0.008*** (0.002)				-0.004** (0.002)		
Exposure Share ²⁰²⁰			-0.004** (0.002)				-0.004** (0.002)	
Exposure Share ²⁰²⁴				-0.001 (0.001)				-0.000 (0.001)
State FE	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓
D.V. Mean	0.566	0.587	0.600	0.619	0.572	0.594	0.606	0.624
N	1139	1139	1139	1137	887	887	887	886
R ²	0.981	0.968	0.987	0.996	0.983	0.970	0.987	0.996

Notes: The pooled sample contains all counties that have produced a Miss USA or a Miss America state titleholder, with an exposure share of zero for counties that produced no Trump-era Miss USA titleholder. The matched sample pairs each Miss America county with a Miss USA county by 1:1 nearest-neighbor propensity-score matching (caliper 0.01) on the covariates listed below (Section 3.6). The dependent variable is the Republican presidential vote share in the indicated election year. Exposure Share^{Year} is constructed according to the demographic structure of the eligible voting population in that year. Controls include an indicator for having produced a Miss USA state titleholder, the previous election's Republican vote share, poverty rate, median household income, employment, annual payroll, number of firms, population, youth ratio, sex ratio, and Black, Asian, and Hispanic population shares. Standard errors are clustered at state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D3. *Miss America Exposure and Vote Shares: Placebo*

	2016 Republican Vote Share		
	(1)	(2)	(3)
Exposure Share	-0.008*** (0.002)		-0.012*** (0.003)
Exposure Share (Miss America)		-0.004 (0.003)	-0.006 (0.004)
State FE	✓	✓	✓
Controls	✓	✓	✓
D.V. Mean	0.556	0.582	0.527
N	707	879	447
R ²	0.969	0.968	0.972

Notes: The dependent variable is 2016 Republican vote share. Exposure Share is the fraction of voting-eligible residents who were aged 10 or older when a Trump-era Miss USA state titleholder from the county was crowned. Exposure Share (Miss America) is the fraction of voting-eligible residents who were aged 10 or older when a Trump-era Miss America state titleholder from the county was crowned. Controls include 2012 Republican vote share, poverty rate, median household income, employment, annual payroll, number of firms, population, youth ratio, sex ratio, Black, Asian and Hispanic proportions. Robust standard errors are clustered at state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D4. Alternative Measures

	2016 Republican Vote Share		
	(1)	(2)	(3)
Log(Exposure Intensity Share + 1)	-0.007*** (0.002)		
# Miss (Trump)		-0.003** (0.001)	
# Miss (AllPeriods)		0.001* (0.000)	
HighExposureCounty			-0.008*** (0.002)
State FE	✓	✓	✓
Controls	✓	✓	✓
D.V. Mean	0.556	0.633	0.556
N	707	3109	707
R ²	0.969	0.966	0.969

Notes: The dependent variable is 2016 Republican vote share. Exposure Intensity Share is the titleholder-weighted fraction of voting-eligible residents who were aged 10 or older when a Trump-era Miss USA state titleholder from the county was crowned. High Exposure County is a dummy indicating that the annual average number of Miss USA state titleholders during Trump's ownership exceeds the all-period average. # Miss (Trump) is the number of Miss USA state titleholders during Trump's era and # Miss (AllPeriods) is the number of Miss USA state titleholders across all periods. Controls include 2012 Republican vote share, poverty rate, median household income, employment, annual payroll, number of firms, population, youth ratio, sex ratio, Black, Asian and Hispanic proportions. Robust standard errors are clustered at state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D5. Election Returns: Alternative Samples

	2016 Republican Vote Share						
	Sample: population percentile				Sample: mobility rate		
	> 5th (1)	> 10th (2)	< 90th (3)	< 95th (4)	< 3% (5)	< 5% (6)	< 10% (7)
Exposure Share	-0.010*** (0.002)	-0.010*** (0.002)	-0.004** (0.002)	-0.005** (0.002)	-0.010*** (0.004)	-0.011*** (0.003)	-0.008*** (0.002)
State FE	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓
D.V. Mean	0.547	0.538	0.580	0.568	0.526	0.555	0.559
N	673	637	634	672	260	518	625
R ²	0.968	0.969	0.969	0.970	0.978	0.971	0.968

Notes: The dependent variable is the 2016 Republican vote share. Columns (1)–(4) exclude counties below the 5th or 10th percentile, or above the 90th or 95th percentile, of the population distribution. Columns (5)–(7) restrict the sample to counties whose mobility rate—the share of residents who recently relocated, from the American Community Survey—is below 3, 5, or 10 percent. Exposure Share is the fraction of voting-eligible residents who were aged 10 or older when a Trump-era Miss USA state titleholder from the county was crowned. Controls include 2012 Republican vote share, poverty rate, median household income, employment, annual payroll, number of firms, population, youth ratio, sex ratio, Black, Asian and Hispanic proportions. Robust standard errors are clustered at state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E. Additional Individual-Level Results

Table E1. Down-ballot placebo tests

	Preference for Voting Republican		
	Congressman	Governor	Senator
	(1)	(2)	(3)
<i>Exposure</i> $\times$ <i>Post</i>	-0.005 (0.013)	0.010 (0.011)	-0.006 (0.015)
Birth Year FE	✓	✓	✓
County $\times$ Survey Date FE	✓	✓	✓
Controls	✓	✓	✓
D.V. Mean	0.254	0.041	0.179
N	20606	20606	20606
R^2	0.550	0.514	0.513

Notes: This table tests for effects of pageant exposure on preferences for other Republican candidates. The dependent variables are dummies indicating immediate vote preference for Republican Congressman, Governor, and Senator, respectively. *Exposure* is the individual-level Miss USA pageant exposure measure defined in Equation (4), and *Post* is an indicator equal to one if the respondent was surveyed after October 7, 2016. Controls include gender, education, race, employment status, party registration, social media usage, and ideological self-placement. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

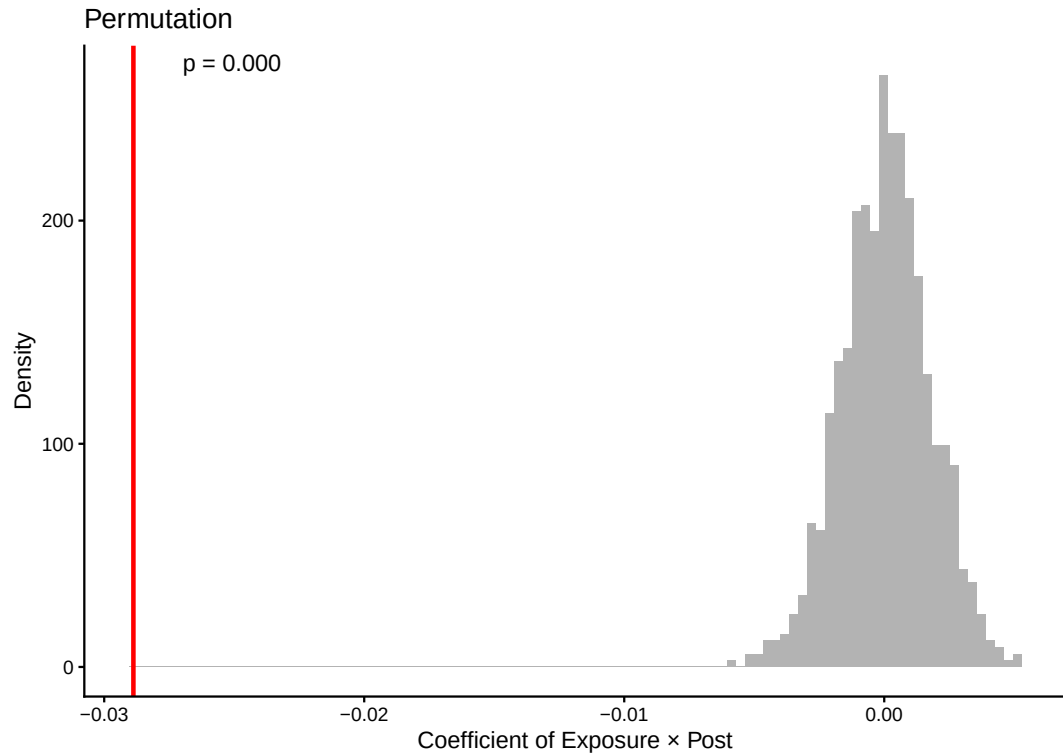


Figure E1. Permutation Test: Individual-Level Estimates

Notes: This figure plots the distribution of the estimated coefficients from the permutation test, where the individual-level exposure measure is randomly reassigned across individuals. The vertical line indicates the observed coefficient from the main specification.

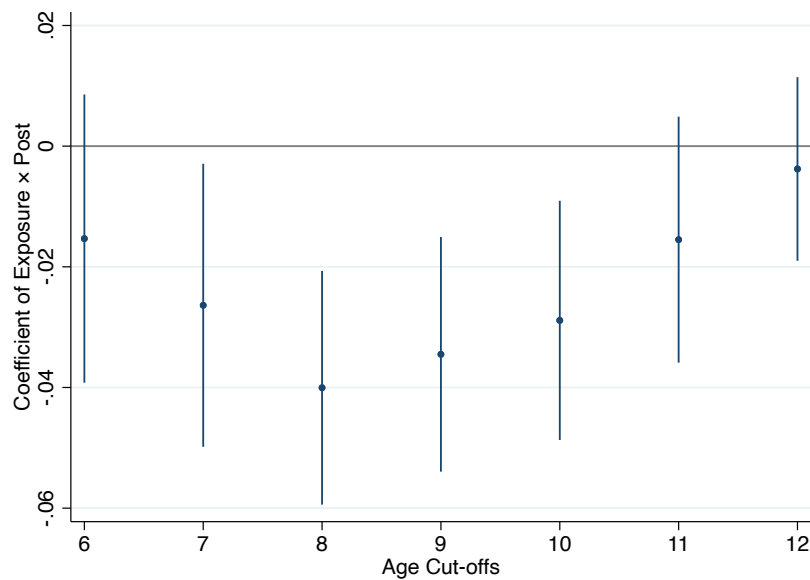


Figure E2. Age Threshold Robustness Check

Notes: This figure plots the coefficients from the age threshold robustness check, where the individual-level exposure measure is constructed using alternative age thresholds between 6 and 12.

Table E2. Heterogeneity by Demographic Characteristics

	Preference for Voting Trump				
	(1)	(2)	(3)	(4)	(5)
<i>Exposure</i> × <i>Post</i>	-0.029*** (0.010)	-0.033*** (0.010)	-0.034*** (0.011)	-0.029*** (0.011)	-0.032*** (0.010)
<i>Exposure</i> × <i>Post</i> × Female	0.001 (0.004)				
<i>Exposure</i> × <i>Post</i> × White		0.005 (0.004)			
<i>Exposure</i> × <i>Post</i> × College			0.006 (0.005)		
<i>Exposure</i> × <i>Post</i> × Employed				-0.000 (0.004)	
<i>Exposure</i> × <i>Post</i> × Elderly					0.004 (0.003)
Birth Year FE	✓	✓	✓	✓	✓
County × Survey Date FE	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓
D.V. Mean	0.318	0.318	0.318	0.318	0.318
N	16613	16613	16613	16613	16613
R ²	0.710	0.710	0.710	0.709	0.710

Notes: The table examines whether the effect of the Access Hollywood tape varies by demographic characteristics. College is a dummy indicating whether she completes college education or above. Employed is a dummy indicating whether she has a full-time job. Elderly is a dummy indicating whether she is over 60 years old. *Exposure* is the individual-level Miss USA pageant exposure measure defined in Equation (4), and *Post* is an indicator equal to one if the respondent was surveyed after October 7, 2016. Controls include all interactions, gender, education, race, employment status, party registration, social media usage, and ideological self-placement. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E3. Main Results Controlling Cohort Dynamic Effects

	Preference for Voting Trump		
	(1)	(2)	(3)
<i>Exposure</i> × <i>Post</i>	-0.029*** (0.010)	-0.029*** (0.010)	-0.039*** (0.015)
Birth Year FE	✓	✓	
County × Survey Date FE	✓	✓	✓
Controls	✓	✓	✓
Survey Date × Birth Year Trend		✓	
Survey Date × Birth Year FE			✓
D.V. Mean	0.318	0.318	0.316
N	16613	16613	16285
R ²	0.709	0.710	0.730

Notes: *Exposure* is the individual-level Miss USA pageant exposure measure defined in Equation (4), and *Post* is an indicator equal to one if the respondent was surveyed after October 7, 2016. Controls include gender, education, race, employment status, party registration, social media usage, and ideological self-placement. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E4. The Effects of Scandal by Political News Consumption

	Intends to vote for Trump, by political news consumption		Political News Consumption
	Yes (1)	No (2)	(3)
<i>Exposure</i> × <i>Post</i>	-0.035*** (0.012)	-0.011 (0.034)	-0.009 (0.014)
Birth Year FE	✓	✓	✓
County × Survey Date FE	✓	✓	✓
Controls	✓	✓	✓
D.V. Mean	0.304	0.327	0.718
N	11469	3466	20606
R ²	0.729	0.731	0.233

Notes: *Exposure* is the individual-level Miss USA pageant exposure measure, and *Post* is an indicator equal to one if the respondent was surveyed after October 7, 2016. Political news consumption is proxied by a CCES question whether the respondent read a story or watched a video about politics. Controls include gender, education, race, employment status, party registration, social media usage, and ideological self-placement. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

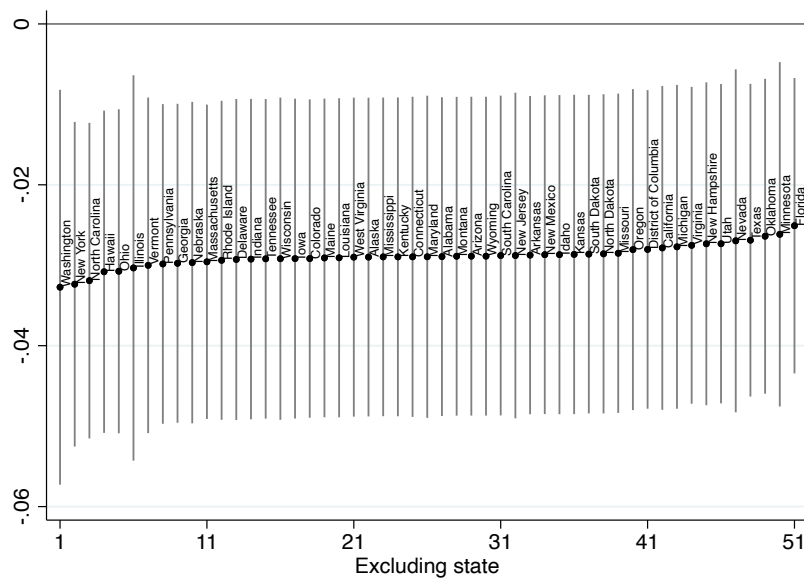


Figure E3. *Leave-One-State-Out Test*

Notes: This figure plots the coefficients from the leave-one-state-out test, where the main specification is re-estimated excluding each state one at a time.

F. Post-Election Votes

Table F1. Individual Miss USA Exposure and Voting Results

	Vote Trump	Vote Romney
	(1)	(2)
Panel A: Full Sample		
<i>Exposure</i>	-0.010** (0.004)	-0.002 (0.005)
D.V. Mean	0.356	0.316
N	32240	32240
Panel B: Counties with Miss USA		
<i>Exposure</i>	-0.009** (0.004)	0.004 (0.006)
Birth Year FE	✓	✓
County FE	✓	✓
Controls	✓	✓
D.V. Mean	0.324	0.294
N	22554	22554

Notes: The dependent variables are indicators for having voted for Trump in the 2016 election and for Romney in the 2012 election. Controls include gender, education, race, employment status, party registration, social media usage, and ideological self-placement. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post-election wave of the 2016 CCES records each respondent's reported vote in the general election. We relate it to the individual exposure measure of Section 4.2 with birth-cohort and county fixed effects and the individual controls of Equation (5):

$$Y_{icb}^{POST} = \alpha + \rho Exposure_{icb} + \phi_b + \theta_c + \Gamma' X_i + \epsilon_{icb}, \quad (15)$$

where Y_{icb}^{POST} indicates a reported vote for Trump in 2016, and standard errors are clustered at the county level. County fixed effects confine identification to cohorts within a county, as in the main individual-level design. Table F1 reports the results. Respondents with greater exposure were about one percentage point less likely, per unit of exposure, to report having voted for Trump in 2016, significant at the 5% level, in the full sample and in the baseline sample of pageant-producing counties (column 1); there is no corresponding relationship for the 2012 Romney vote (column 2). The pattern matches the county-level estimates of Section 5.3 in sign and in magnitude.

G. Derivations and Proofs

Solution of the voter's problem. For $a > 0$ the signal $s = \theta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2/a)$ is Gaussian, so the posterior of θ is normal with precision $h_0 + a/\sigma^2$. The estimator that minimizes expected squared error is the posterior mean, $\hat{\theta} = \mu_0 + \kappa(a)(s - \mu_0)$ with $\kappa(a) = (a/\sigma^2)/(h_0 + a/\sigma^2)$, and the minimized error is the posterior variance, $1/(h_0 + a/\sigma^2)$. Problem (10) therefore reduces to

$$\min_{a \geq 0} V(a) \equiv \frac{1}{h_0 + a/\sigma^2} + \psi_i(n) a,$$

with $V'(a) = -\sigma^{-2}(h_0 + a/\sigma^2)^{-2} + \psi_i(n)$ and $V''(a) = 2\sigma^{-4}(h_0 + a/\sigma^2)^{-3} > 0$, so V is strictly convex. The first-order condition gives $(h_0 + a/\sigma^2)^2 = 1/(\sigma^2\psi_i(n))$, that is, $h_0 + a/\sigma^2 = 1/(\sigma\sqrt{\psi_i(n)})$, whose solution is $a = \sigma/\sqrt{\psi_i(n)} - \sigma^2 h_0$. This is positive if and only if $V'(0) = -\sigma_0^4/\sigma^2 + \psi_i(n) < 0$, that is, $\psi_i(n) < \sigma_0^4/\sigma^2$; otherwise the corner $a = 0$ is optimal. This is (11). At an interior optimum, $\kappa = 1 - h_0/(h_0 + a^*/\sigma^2) = 1 - h_0\sigma\sqrt{\psi_i(n)} = 1 - \sigma\sqrt{\psi_i(n)}/\sigma_0^2$, which is the gain in (12); at the corner the belief is μ_0 . Since $\mathbb{E}[s - \mu_0 \mid \theta] = \theta - \mu_0$, the expected revision is $\mathbb{E}[\hat{\theta} - \mu_0 \mid \theta] = \kappa(\theta - \mu_0)$.

Properties of the connection. Write $A = A_i(n) = \sum_k S(x_n, e_k, t_k)$ and $K = K_i$ in (8), so that $c = A/(A + \rho K)$. Then $\partial c/\partial A = \rho K/(A + \rho K)^2 > 0$ and $\partial c/\partial K = -\rho A/(A + \rho K)^2 \leq 0$. For an episode k related to the cue, $|x_n \cap e_k| > 0$, write $\tau_k = t - t_k$ for its age; then $\partial A/\partial \tau_k = |x_n \cap e_k| g'(\tau_k) \leq 0$, so $\partial c/\partial \tau_k \leq 0$: aging any related episode lowers the connection, strictly where $g' < 0$. Adding an episode that shares features with the cue raises A and K by one each and raises c if and only if its similarity exceeds A/K , the average similarity of what is already in memory; adding an episode that shares no feature with the cue raises K alone and lowers c . The last property is what makes familiarity through channels unrelated to the pageant predict a smaller connection to the tape, as noted at the end of Section 7.

Proof of Proposition 1. Substitute $\psi(c) = \psi/(1 + \phi c)$ from (9) into (11) and (12). (i) The voter attends if and only if $\psi(c) < \sigma_0^4/\sigma^2$, that is, $1 + \phi c > \psi\sigma^2/\sigma_0^4$, that is, $c > \bar{c} = \phi^{-1}(\psi\sigma^2/\sigma_0^4 - 1)$. Since $\partial \bar{c}/\partial \sigma_0^2 = -2\psi\sigma^2/(\phi\sigma_0^6) < 0$ and $\partial \bar{c}/\partial \phi = -\bar{c}/\phi$, the threshold is decreasing in σ_0^2 , and decreasing in ϕ whenever it is positive; it is negative when $\sigma_0^4 > \psi\sigma^2$, in which case every voter attends. (ii) Conditional on attending, $a^*(c) = \sigma\sqrt{(1 + \phi c)/\psi} - \sigma^2 h_0$, and $\partial a^*/\partial c = \sigma\phi/(2\sqrt{\psi(1 + \phi c)}) > 0$. (iii) Conditional on attending, $\kappa(c) = 1 - \sigma\sqrt{\psi}/(\sigma_0^2\sqrt{1 + \phi c})$, and $\kappa'(c) = \sigma\sqrt{\psi}\phi/(2\sigma_0^2(1 + \phi c)^{3/2}) > 0$. Differentiating $\mathbb{E}[\hat{\theta} - \mu_0 \mid \theta] = \kappa(c)(\theta - \mu_0)$ with respect to c gives (13), which is negative for $\theta < \mu_0$. (iv) With a single related episode, $A = |x_n \cap e_k| g(\tau)$, and by the properties above $\partial c/\partial \tau \leq 0$. By the chain rule, $\partial a^*/\partial \tau = (\partial a^*/\partial c)(\partial c/\partial \tau) \leq 0$ and $\partial \kappa/\partial \tau = \kappa'(c)(\partial c/\partial \tau) \leq 0$ conditional on attending, and (14) multiplies two non-

positive terms by $\theta - \mu_0 < 0$, with strict inequality where $g' < 0$. If τ is large enough that $c(\tau) \leq \bar{c}$, part (i) gives $a^* = 0$ and $\kappa = 0$. $\square$